The APL block model, built step by step

> **QUESTION**
>
> How is the APL block model built, step by step, and what does it predict about blocking APL, before any comparison with the TNT / TNT3c recordings?

INPUTS

- All in `with_outside_ROI_synapses/claws_v2m2/` (corrected calyx, new claws; Reports 02 and 04):
- KC claws `hemibrain_KC_claws_derived.csv.gz`, `malecns_KC_claws_derived.csv.gz`; per-KC calyx APL synapse counts `KC_weights_APL_PN_per_KC.csv`; per-glomerulus APL wiring `odor_weighted/APL_wiring_by_glomerulus.csv`
- ORN responses: `orn_inputs_2026-11-03/ORN_TNT16_flymean.csv` (new table from Yang's `responses_wide.csv`; Report 03)
- GH146 means `odor_weighted/GH146_mean_by_glomerulus.csv` (Report 03)
- For Section 13 only: the TNT and TNT3c recordings (Report 06)

OUTPUTS

- `with_outside_ROI_synapses/claws_v2m2/odor_weighted/`: `APL_block_model_per_odor.csv`, `APL_block_model_odor_pairs.csv`, `APL_block_model_other_weights_*.csv`, `APL_block_model_summary.json`, `APL_block_model.png`

SCRIPTS

- `scripts/apl_block_model.py` (the model)
- `scripts/reports/r05_block_model.py` (this report; it re-runs the steps for illustration and checks them against the saved tables)

← Report 04 · All reports · Report 06 →

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This report explains the KC model that all later steps use. Report 07 takes the same model and pins it to the recordings. Here it runs in its **prediction** form, set up before the recordings arrived: 20% of KCs respond to an odor (averaged over odors) with APL blocked, and 10% with APL intact. Every symbol is defined in Table 1, and again where it first appears.

What changed in this revision (2026-11-04)

Built on the corrected calyx (synapses outside the official ROIs added) and the new claws (8 µm along the KC skeleton, one claw per PN with at least 2 PSDs; Report 02), and the new ORN table from `responses_wide.csv` (Report 03).

New inputs: ORN responses passed through a PN-level transformation (Olsen et al. 2010, as used by Kennedy 2019), compared with GH146 (Section 4); ORN and GH146 with their negative responses kept (Section 4). Their results are in Section 11.

New version MC: APL inhibits both the KCs (per KC, as in C) and the PN boutons (per glomerulus, as in P), at the boutons' real share of APL's synapses (Section 9).

Why the input weights are divided by the KC's total (Section 3), with two alternatives run in Section 12; why p_g and q_g are counted per synapse (Section 9, Tables 7 and 8).

ms is kept everywhere; similarities that cannot be computed (an odor that drives no KC) are marked, not dropped.

Every number in the Results is spelled out (Table 9; "r" is always Pearson, "ρ" Spearman), the similarity maps are shown for every version with the pairs that change most and least outlined, and the recordings' similarity maps are added at the end (Section 13).

The model in one paragraph

Each KC adds up its claws' glomerular inputs (a weighted average of the glomeruli's odor responses, with weights from the connectome). It responds by how much that sum exceeds a threshold. APL is modelled as a single unit with one activity level per odor. Every KC excites APL, so this activity is the mean response of all KCs to that odor. APL then inhibits every KC equally: it subtracts its activity times a gain a from each KC's input. The gain is the same for every odor, but the inhibition is not, because APL's activity is higher for odors that drive more KCs. Because APL's activity depends on the KC responses and the KC responses depend on APL's activity, both are solved together for each odor ("self-consistently"). "Blocked" (TNT) means no APL at all. Later versions replace the uniform APL with connectome measurements of where APL's synapses are.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
k	Index of a Kenyon cell (KC) in one mushroom body	$k = 1 \dots N$
N	Number of KCs in the mushroom body (KCs with at least one claw)	hemibrain R 1,771; male CNS L 1,879; male CNS R 1,901
g	Index of an antennal-lobe glomerulus (uniglomerular PN types only)	names such as DM2, VA1v
o	Index of an odor	16 odors (ORN table) or 13 (GH146)
c	Index of a claw (one PN's synapses on one KC within one 8 µm cluster, Report 02)	median 6 claws per KC (hemibrain), 6 (male CNS)
n_c	Number of PN→KC synapses on claw c	synapse count
$g(c)$	Glomerulus of claw c: the glomerulus of its PN (each claw has one PN)	

Symbol	What it stands for	Values, units, range
W_{kg}	Input weight: share of KC k's claw PN synapses that sit on claws of glomerulus g	0 to 1; a row sums to the share from uniglomerular PNs (0.95–0.96 on average)
r_{go}	Response of glomerulus g to odor o; negative values set to 0 (except in the "neg" inputs of Section 4)	ORN table: fly-averaged response as provided (units not stated), −0.22 to 1.63; GH146: mean peak ΔF/F
x_{ko}	Drive of KC k by odor o: weighted average of its glomeruli's responses	same units as r_{go}
θ	KC threshold, one number per mushroom body and input table	set so 20% of KCs respond with APL blocked (Section 5)
y_{ko}	Response of KC k to odor o (how far the drive exceeds threshold plus inhibition; 0 if below)	same units as r_{go}; ≥ 0
A_o	APL activity for odor o: the (weighted) mean KC response	same units as y; one number per odor
a	Subtractive APL gain (how strongly APL activity inhibits a KC), one number per mushroom body and version	dimensionless; set by calibration
b	Divisive APL gain on PN boutons (versions P0, P, L)	dimensionless; set by calibration
u_k	Per-KC inhibition weight: KC k's APL→KC synapses in the calyx ÷ the mean over KCs	mean 1; $u_k = 1$ in the uniform version
v_k	Per-KC drive weight: KC k's KC→APL synapses in the calyx ÷ the mean over KCs	mean 1; $v_k = 1$ in the uniform version
p_g	Bouton weight of glomerulus g: APL→PN synapses per PN→KC synapse of its PNs, scaled to a claw-input-weighted mean of 1	version P (Report 04)
q_g	Claw weight of glomerulus g: APL→KC synapses on its claws per PN→KC synapse on them, scaled the same way	version L (Report 04)
f	Boutons' share of APL's calyx output synapses: APL→PN ÷ (APL→PN + APL→KC)	hemibrain 0.1, male CNS 0.103
s	Common scale of the KC and bouton gains in versions M0, M and MC	set by calibration
$F_o(A)$	The APL activity the KCs would produce for odor o if APL's activity were A	used to solve for A_o
ϕ_o	Fraction of KCs responding to odor o ($y_{ko} > 0$), also called the coding level	0 to 1
S_{ab}	Cosine similarity of the KC population responses to odors a and b	0 to 1 here (responses are ≥ 0)
ΔS_{ab}	Effect of blocking APL on the similarity of odors a and b: S_{ab} blocked − S_{ab} intact (written Δcos in the figure files)	positive = more similar when blocked
w_g	Total claw input from glomerulus g in the mushroom body, $\sum_k W_{kg}$	used to scale glomerulus weights

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1 What the model is

The model is a rate model of the calyx. A rate model describes each cell by one number per odor (its response), not by spikes or a time course. There is one model per mushroom body: hemibrain R, male

CNS L and male CNS R. FlyWire and BANC were dropped on 2026-10-21 at Yang's request. Figure 1 shows the circuit and where each symbol acts.

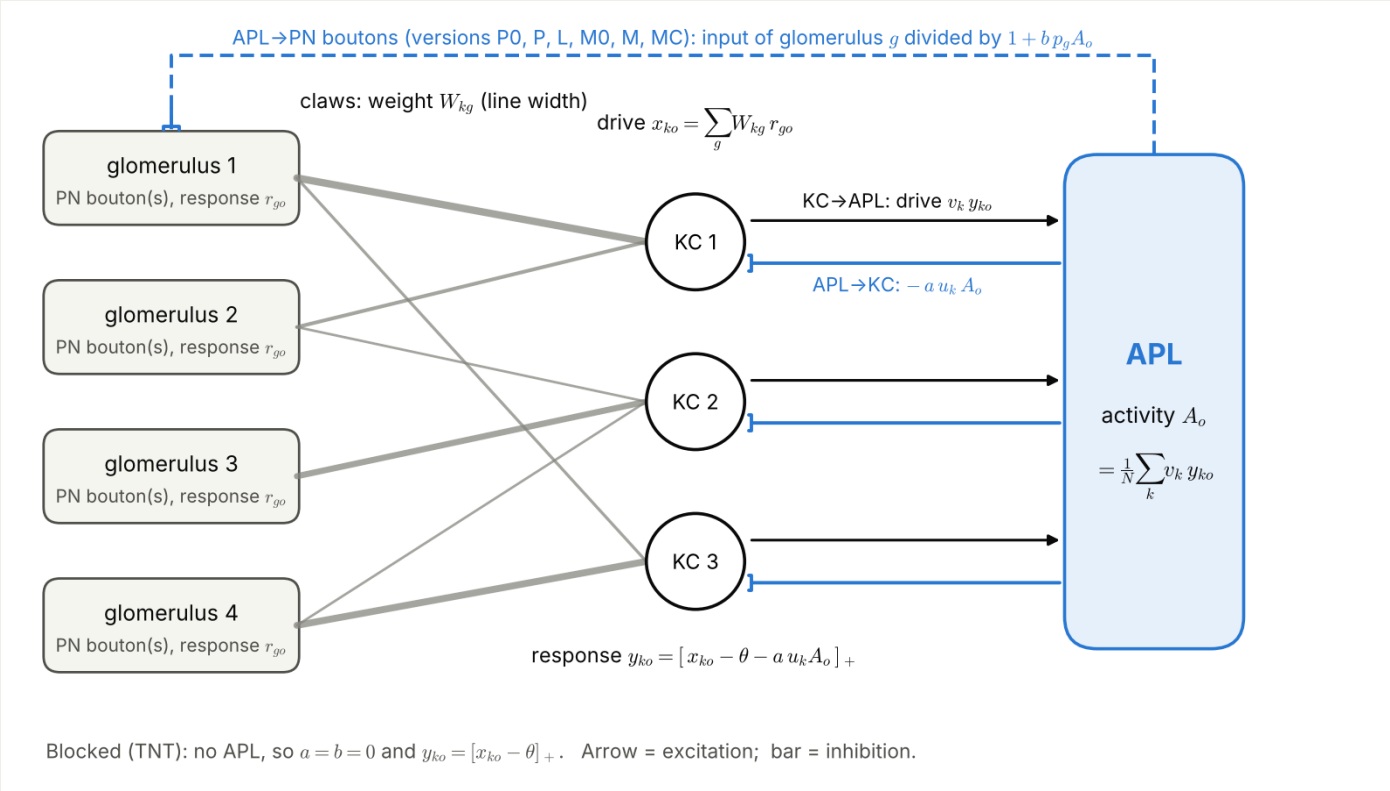


Figure 1. The circuit in the model and where each symbol acts. Glomeruli (left) send PN input to KCs (middle) through claws. Line width shows the input weight W_{kg}, the share of a KC's claw PN synapses that come from each glomerulus (three example KCs). A KC's drive x_{ko} is the W-weighted sum of its glomeruli's responses r_{go}. KCs excite APL (black arrows), and APL's activity A_o is their mean response, each KC weighted by v_k. APL inhibits each KC by subtracting $a\,u_k\,A_o$ (blue bars). In the bouton versions it instead divides the input of glomerulus g by $1 + b\,p_g\,A_o$ (dashed blue path). In the uniform version every u_k, v_k and p_g is 1.

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2 Step 1: one model per mushroom body

The hemibrain contains one calyx, the right one (1,771 KCs with claws). The male CNS contains both. Its KCs are split into left and right in two steps. First, each KC's position is the mean of its claw centroids. Second, two-means clustering (k-means with $k = 2$, seed 0) on those positions gives two groups. Each group is named after the APL that its KCs' on-claw APL synapses mostly contact (APL_L or APL_R). This gives 1,879 KCs on the left and 1,901 on the right (Figure 2). Each mushroom body gets its own W, u, v and threshold, and its own calibrated gains. Results are averaged over the three only at the end.

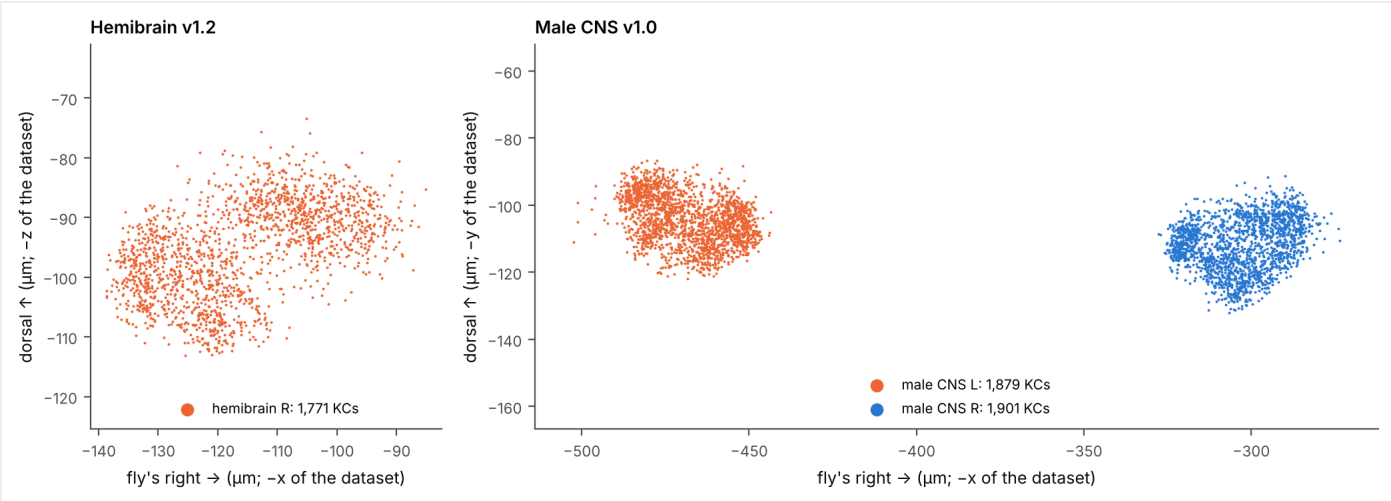


Figure 2. KCs of each mushroom body. Each dot is one KC, placed at the mean of its claw centroids and seen from behind the fly (posterior view: the fly's right on the right, dorsal up). Axes are the dataset's own coordinates in µm, sign-flipped for this view: hemibrain (−x, −z), male CNS (−x, −y). Color shows the mushroom body each KC is assigned to. In the male CNS the two groups come from two-means clustering and are named by the APL their on-claw synapses contact.

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3 Step 2: input weights from the claws

A **claw** is a group of one KC's postsynaptic sites from one antennal-lobe PN in the calyx (Report 02). The KC's calyx synapse sites are first clustered along its skeleton: sites within 8 µm of each other, measured along the KC's branches, belong to the same cluster (single linkage). Within a cluster, every PN that makes at least 2 synapses onto the KC gets its own claw. So each claw has exactly one PN and one glomerulus, $g(c)$, and PNs that make a single synapse in a cluster are not counted. Only uniglomerular PN types give input. Claws of multiglomerular, bi-glomerular or untyped PNs stay in the denominator but give no input. The input weight is

$$W_{kg} = \frac{\sum_{c \in \text{claws}(k):\, g(c)=g} n_c}{\sum_{c \in \text{claws}(k)} n_c} \tag{1}$$

n_c = PN synapses on claw c; the sum on top runs over KC k's claws of glomerulus g, the sum below over all its claws

Worked example. Table 2 follows one hemibrain αβ KC (body ID 1016122913, type KCab-c) through Equations 1 and 2 for the odor pentyl acetate (pa). Its 6 claws carry 36 PN synapses in total.

Table 2. One KC's claws, its input weights and its drive by pentyl acetate. Each row is a claw. Its share of the KC's claw PN synapses is its contribution to W_{kg}; claws of the same glomerulus add up. The last column is the claw's contribution to the drive. Glomeruli missing from the ORN table count as 0.

claw	main PN type	glomerulus $g(c)$	PN synapses n_c	share = contribution to W_{kg}	$r_{g,\text{pa}}$ (ORN, ≥ 0)	$W \times r$
claw 1	VC3m_lvPN	VC5	15	0.417	not in table (0)	0.000
claw 2	VL2p_adPN	VL2p	6	0.167	0.098	0.016
claw 3	DC1_adPN	DC1	5	0.139	0.164	0.023
claw 4	VA3_adPN	VA3	4	0.111	0.135	0.015
claw 5	VA1v_adPN	VA1v	4	0.111	not in table (0)	0.000

claw	main PN type	glomerulus $g(c)$	PN synapses n_c	share = contribution to W_{kg}	$r_{g,\mathrm{pa}}$ (ORN, ≥ 0)	$W \times r$
claw 6	DM5_lPN	DM5	2	0.056	0.152	0.008
sum			36	1.000		**0.063** = $x_{k,\mathrm{pa}}$

The **drive** of KC k by odor o is the W-weighted sum of its glomeruli's responses:

$$x_{ko} \;=\; \sum_{g} W_{kg}\; r_{go} \tag{2}$$

Because the weights of one KC add up to at most 1, x_{ko} is a weighted average of the responses of the KC's glomeruli. Claws of glomeruli outside the response table count as 0. The drive is therefore smaller for KCs whose claws come mostly from unmeasured glomeruli.

3.1 Why W_{kg} is divided by all of the KC's claw PN synapses

Dividing by the KC's total makes its drive a **weighted average** of its glomeruli's responses, so every KC's drive is on the same scale whatever its size. Without it, the drive would be $\sum_g n_{kg}\, r_{go}$ (n_{kg} = the KC's claw PN synapses from glomerulus g), which grows with the number of synapses. KCs differ a lot in that number: hemibrain KCs have a median of 91 claw PN synapses (5–95%: 36–196), and γ KCs have about 2.1 times as many as the others (159 on average, against 75 for αβ and 73 for α′β′; male CNS: 132, 89 and 79). With raw counts and one threshold, γ KCs would take most of the responses (Section 12 shows this). The averaging assumes that a KC with more input synapses also needs more input to reach threshold, which is roughly what a larger dendrite (lower input resistance) does.

The denominator also includes claws of PNs that are not in the response table or are multiglomerular. Those claws count as input 0, so a KC whose claws come mostly from unmeasured glomeruli gets a smaller drive, instead of having its few measured claws scaled up.

This is a modelling choice, and the data do not settle it. Section 12 runs two alternatives: raw synapse counts ($W_{kg} = n_{kg}$, one threshold), and equal weight per claw (W_{kg} = share of the KC's claws that come from g).

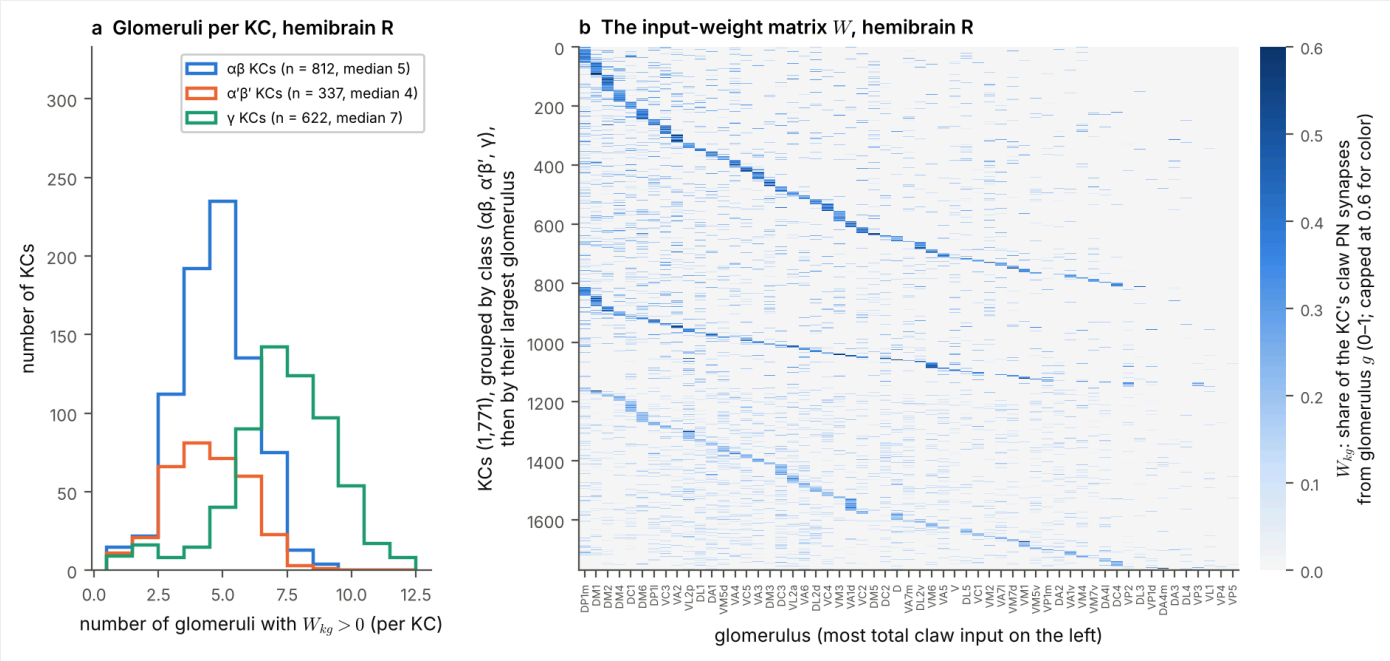


Figure 3. Input weights W_{kg} across all KCs. a, For each KC, the number of glomeruli that give it any input ($W_{kg} > 0$), as a histogram per KC class. **b**, The whole matrix W for hemibrain R. Each row is a KC and each column a glomerulus; color is W_{kg} (white = 0). Values above 0.6 share the darkest color. Every KC samples a few glomeruli, and different KCs sample different combinations.

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4 Step 3: odor input to the glomeruli

Two tables give r_{go}, the response of glomerulus g to odor o. In the main runs these numbers go straight into Equation 2, and negative responses (inhibition below baseline) are set to 0. Two variations follow below: a PN-level transformation of the ORN responses, and keeping the negative values.

- **ORN table** (`orn_inputs_2026-11-03/ORN_TNT16_flymean.csv`, built from Yang's `responses_wide.csv`; Report 03): 38 glomeruli × 16 odors, fly-averaged olfactory receptor neuron (ORN) responses. 12 odors come from the megamat flies and 4 (paa, -aPine, euc, B-myr) from the validation2 flies. DL4 and VM4 were imaged only in the validation2 flies, so they have values for 4 of the 16 odors and are left out (their claws give input 0). The file does not state units (range −0.22 to 1.63). The values are about 0.7–0.8 times those of the old table, with the same patterns (Report 03); the shared threshold absorbs a common scale. Glomeruli in this table carry 78%–79% of the KCs' claw PN synapses (mean over KCs, per mushroom body).
- **GH146 table**: mean peak ΔF/F of GH146 PNs (7 flies), 29 glomeruli × 13 odors. -aPine, euc and B-myr are missing, and paa comes from the diagnostics panel at 10^{-5} because it is the only PAA presentation. These glomeruli carry 63%–64% of the claw PN synapses.

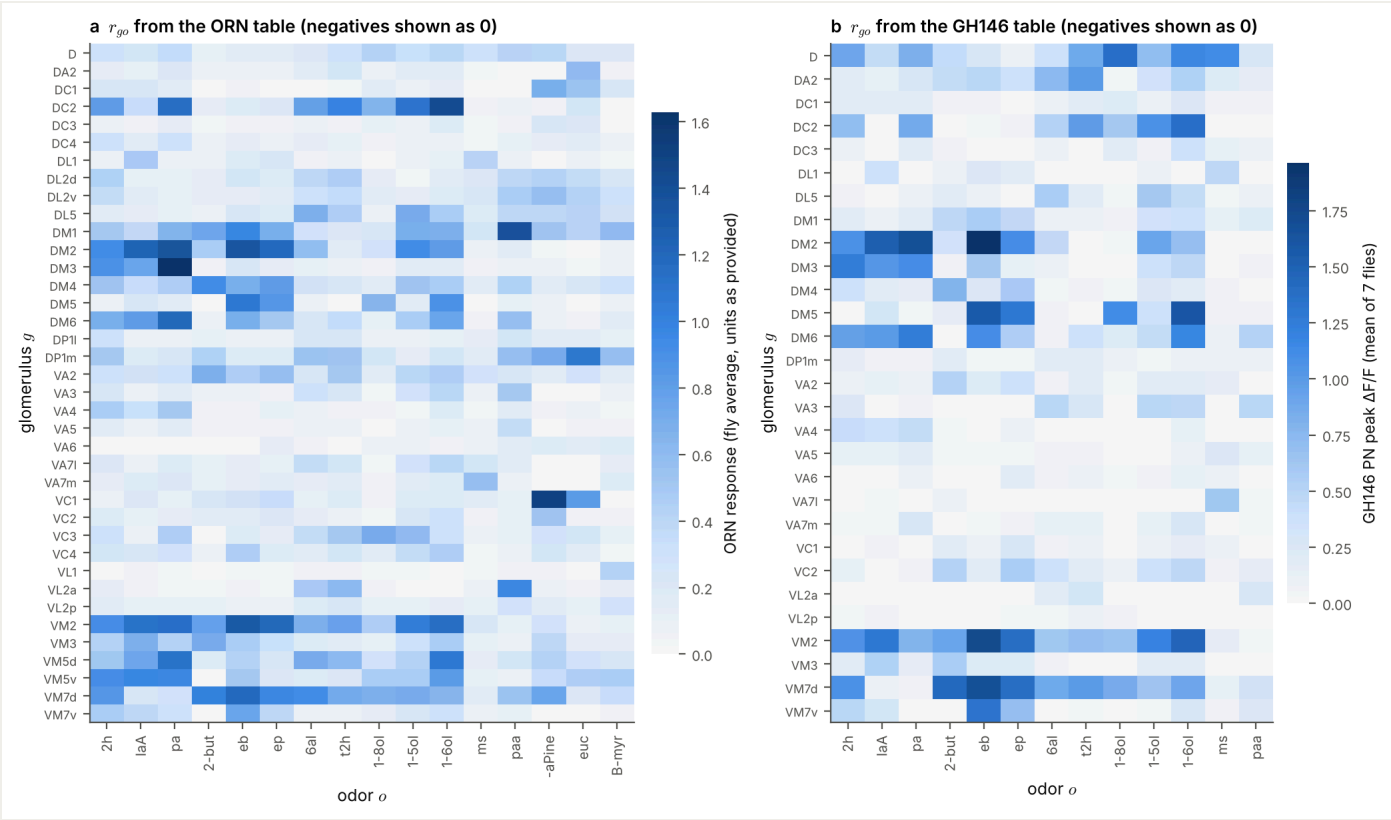


Figure 4. The glomerular responses r_{go} that drive the model. Each cell is one glomerulus's response to one odor after negative values are set to 0. Color runs from 0 (white) to the table's maximum (dark blue), with a separate scale for each panel. **a**, ORN table, 38 glomeruli × 16 odors. **b**, GH146 table, 29 glomeruli × 13 odors. Glomeruli that are missing here give 0 input to the claws that belong to them.

4.1 A PN-level transformation of the ORN responses

KCs receive input from PNs, not ORNs. Olsen, Bhandawat and Wilson (2010) showed that a PN's response is a saturating function of its own ORNs' input, divided down by the summed ORN input to all glomeruli (lateral inhibition in the antennal lobe). Kennedy (2019, "Learning with naturalistic odor representations in a dynamic model of the Drosophila olfactory system") used the same function to turn ORN responses into PN input to KCs. Without its maximum rate $R_{\max}$, which only scales all values and is absorbed by the threshold, it is

$$r_{go}^{\mathrm{PN}} = \frac{r_{go}^{1.5}}{r_{go}^{1.5} + \sigma^{1.5} + (m\, s_o)^{1.5}}, \qquad s_o = \frac{1}{G}\sum_{g} r_{go} \tag{3}$$

r_{go} = ORN response (negative values set to 0); s_o = mean ORN response over the G measured glomeruli for odor o; σ = half-saturation constant; m = strength of the lateral inhibition

Olsen's σ and m were fitted to spike rates and do not carry over to these calcium responses. So both were fitted here: σ from the 2nd percentile of the positive ORN responses up to 10 times the largest one, and m from 0 to 16, keeping the pair whose PN responses best match GH146 (Pearson r over the 29 glomeruli × 13 odors that both tables share). The best fit is σ = 2.44 and m = 0. With m = 0 there is no lateral inhibition, and with σ larger than every positive ORN response (the largest is 1.63) the function is close to a power law, $r^{1.5}$, with only the largest responses starting to saturate. Its match with GH146 is **r = 0.825**, against **r = 0.821** for the untransformed ORN responses: practically the same. Strong saturation (small σ) matches GH146 clearly worse, and strong lateral inhibition (large m) slightly worse (Figure 5c). So on these data a PN-level transformation of this form barely brings the ORN input closer to the measured PN responses. The model is run with the best-fitting PN responses as the input "ORN-PN" (Section 11).

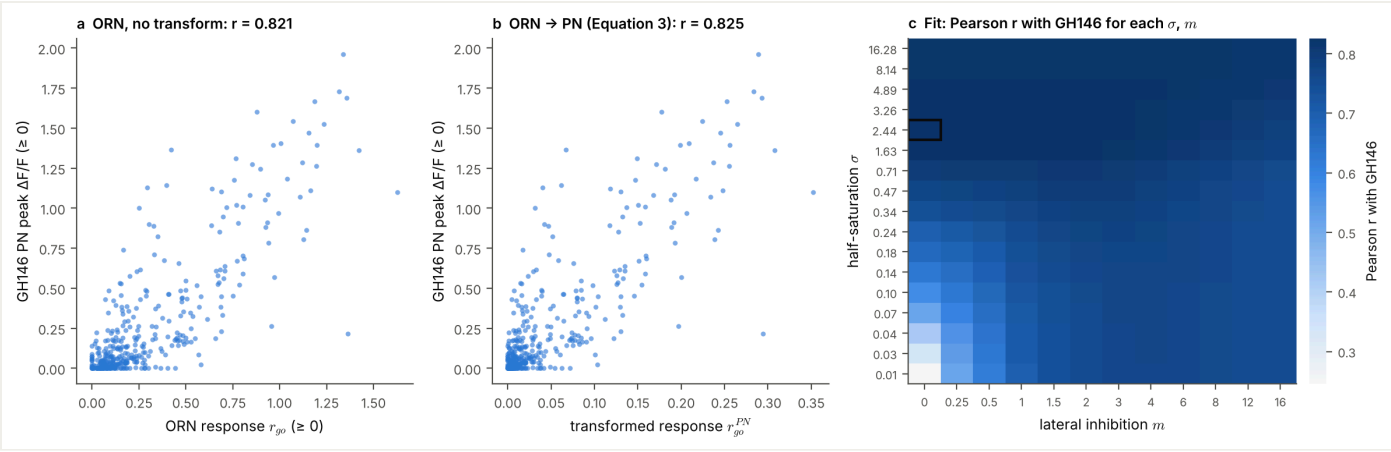


Figure 5. ORN → PN transformation compared with the GH146 responses. Each point in **a** and **b** is one glomerulus–odor value (29 glomeruli × 13 odors shared by the ORN and GH146 tables). **a**, GH146 against the ORN responses as they are. **b**, GH146 against the ORN responses after the PN transformation at the best σ and m. **c**, The Pearson r with GH146 over the grid of σ (rows; the 2nd to 90th percentiles of the positive ORN responses, then 1 to 10 times the largest response) and m (columns); the black square is the best pair.

4.2 Negative responses

In the main runs, responses below baseline are set to 0. They are few and small: 27 of 608 ORN values (lowest −0.22, against a maximum of 1.63) and 57 of 377 GH146 values (lowest −0.16, maximum 1.96). The most negative are VA7m–-aPine (−0.22), VA7m–euc (−0.10), VA7l–-aPine (−0.08) (ORN) and VA7l–1-6ol (−0.16), VA7l–1-8ol (−0.12), DL1–2h (−0.10) (GH146).

The inputs "ORN neg" and "GH146 neg" keep them. A negative response then lowers the drive of every KC with claws from that glomerulus (Equation 2 is unchanged), and θ and the gains are recalibrated as before. One side effect concerns the versions that divide bouton input (P0, P, L, M0, M, MC): dividing a negative input by $1 + b\,p_g A_o$ makes it less negative, so in those versions APL also weakens the inhibition that a glomerulus passes on. The results are in Section 11.

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5 Step 4: KC drive and the threshold with APL blocked

With APL blocked there is no inhibition, so a KC responds by how far its drive exceeds the threshold θ:

$$y_{ko}^{\text{blocked}} = [\,x_{ko} - \theta\,]_+ \qquad \text{where } [z]_+ = \max(z, 0) \tag{4}$$

θ is one number per mushroom body and input table. It is found by bisection so that, averaged over odors, 20% of KCs respond ($y_{ko} > 0$). The bisection keeps an interval $[\theta_{lo}, \theta_{hi}]$, which starts at 0 and the largest drive. At each step it tests the midpoint: if more than 20% respond there, the threshold must be higher, so the midpoint becomes θ_{lo}; otherwise it becomes θ_{hi}. 60 halvings make the interval negligibly small. Individual odors then respond at different levels (Figure 6). Strong odors drive many KCs past the shared threshold, and weak ones drive few.

$$\frac{1}{O}\sum_{o=1}^{O} \phi_o^{\text{blocked}} = 0.20, \qquad \phi_o = \frac{1}{N}\#\{k : y_{ko} > 0\} \tag{5}$$

O = number of odors; ϕ_o = fraction of KCs responding to odor o

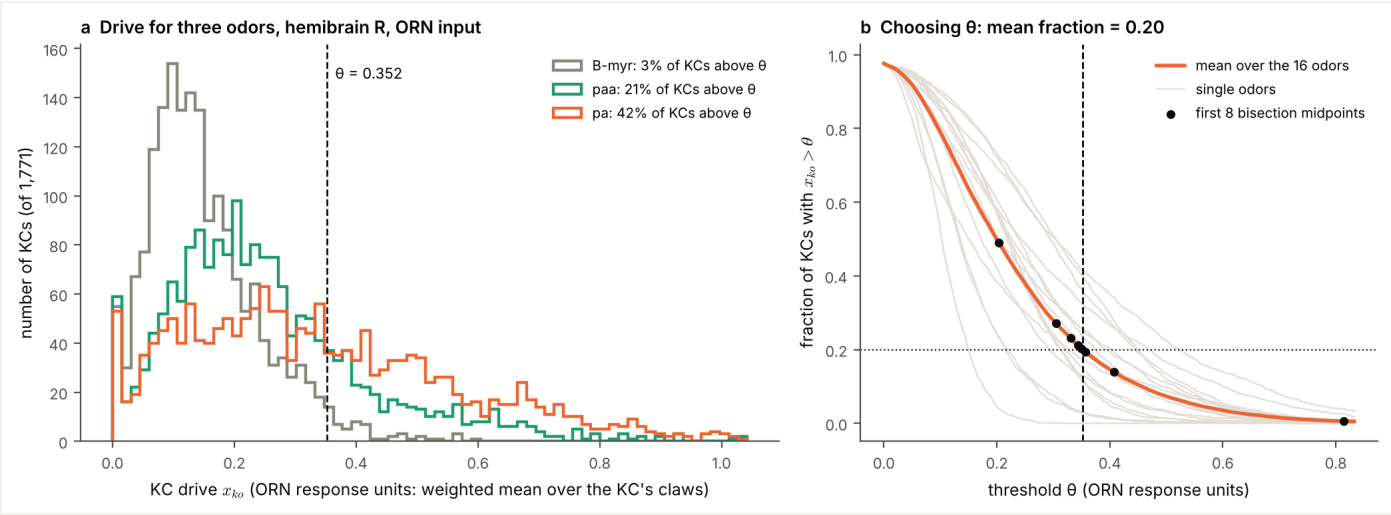


Figure 6. The KC drive and how the threshold θ is set. a, Histograms of the drive x_{ko} over the 1,771 hemibrain KCs for a weak, a middle and the strongest odor (ORN input). The dashed line is the threshold θ. KCs to its right respond when APL is blocked. **b**, The fraction of KCs above a threshold, as the threshold rises: thin grey lines are single odors, and the orange line is their mean. θ is where the orange line crosses 0.20 (dotted). Black dots are the first bisection midpoints as they close in on it.

Table 3. Thresholds. In each case θ makes 20% of KCs respond with APL blocked, averaged over odors. The units are those of the response table.

mushroom body	θ with ORN input	θ with GH146 input
hemibrain R	0.352	0.273
male CNS L	0.366	0.274
male CNS R	0.369	0.287

6 Step 5: APL feedback and what "solved self-consistently" means

With APL intact, each KC is inhibited in proportion to APL's activity for that odor:

$$y_{ko} \;=\; [\,x_{ko} - \theta - a\,u_k\,A_o\,]_+ \tag{6}$$

and APL's activity is the mean response of the KCs, each weighted by how strongly it drives APL:

$$A_o \;=\; \frac{1}{N}\sum_{k=1}^{N} v_k\,y_{ko} \tag{7}$$

In the uniform version (U) every $u_k = v_k = 1$. Every KC then gets the same inhibition aA_o for a given odor, and APL's activity is simply the mean KC response. The gain a is one number per mushroom body and version. It is the same for every odor and every KC. The inhibition still differs between odors, because A_o does.

Why "self-consistently"

A_o appears on both sides. The KC responses (Equation 6) depend on APL's activity, and APL's activity (Equation 7) depends on the KC responses. The model needs the one value of A_o that reproduces itself: the value that, when used to compute the KC responses, gives back the same A_o as their mean. Mathematically this is a fixed point. Physically, it is the steady state in which APL and the KCs have settled.

To find it, define for each odor the function

$$F_o(A) \;=\; \frac{1}{N}\sum_k v_k\,[\,x_{ko} - \theta - a\,u_k\,A\,]_+ \tag{8}$$

the APL activity the KCs would produce if APL's activity were A

and solve $A = F_o(A)$. F_o starts at $F_o(0)$, the mean (weighted) response with no inhibition, and can only fall as A grows, because more inhibition never raises a KC's response. The line A rises from 0. The two cross exactly once, which makes the solution unique (Figure 7). The code finds it by bisection on $[0, F_o(0)]$: if $A > F_o(A)$ at the midpoint, the crossing lies below it; otherwise above. 60 halvings, done for all odors at once (function `fixed_point` in `apl_block_model.py`).

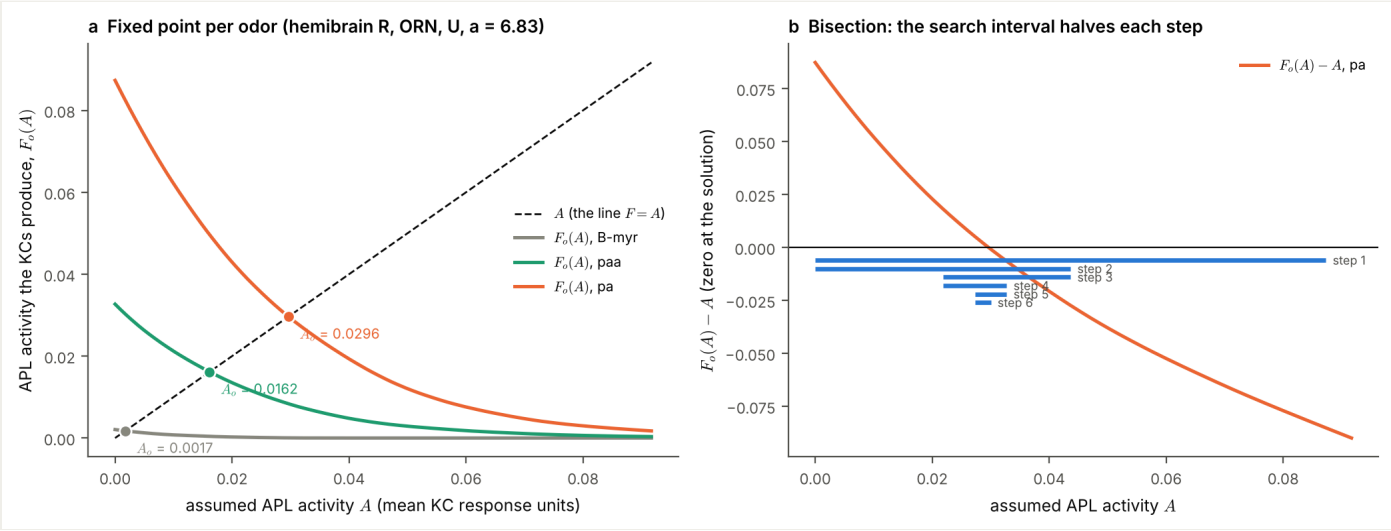


Figure 7. Solving for APL's activity: the fixed point $A_o = F_o(A_o)$. a, For three odors (a weak one, a middle one and the strongest; hemibrain R, ORN input, uniform APL at its calibrated gain), the APL activity that the KC responses would produce, $F_o(A)$ (colored), against the assumed activity A. The dashed diagonal is $F = A$, and the dots mark where each curve crosses it: the self-consistent APL activity A_o. Stronger odors cross further out, so they recruit more APL. **b**, The bisection for the strongest odor. Blue bars are the search interval $[A_{lo}, A_{hi}]$ at each step (drawn below zero for visibility). Each step halves the interval around the zero of $F_o(A) - A$.

Table 4. The first bisection steps for pa (hemibrain R, ORN input, uniform APL). The interval starts at $[0, F_o(0)]$. After 60 steps it has shrunk by a factor of 2^{60}. The final A_o is 0.029640.

step	A_{lo}	A_{hi}	midpoint m	$F_o(m)$	decision
1	0.00000	0.08732	0.04366	0.01630	m > F(m): solution below m, so $A_{hi} = m$
2	0.00000	0.04366	0.02183	0.04006	m ≤ F(m): solution above m, so $A_{lo} = m$
3	0.02183	0.04366	0.03275	0.02625	m > F(m): solution below m, so $A_{hi} = m$
4	0.02183	0.03275	0.02729	0.03249	m ≤ F(m): solution above m, so $A_{lo} = m$
5	0.02729	0.03275	0.03002	0.02921	m > F(m): solution below m, so $A_{hi} = m$
6	0.02729	0.03002	0.02865	0.03081	m ≤ F(m): solution above m, so $A_{lo} = m$
7	0.02865	0.03002	0.02934	0.03000	m ≤ F(m): solution above m, so $A_{lo} = m$
8	0.02934	0.03002	0.02968	0.02960	m > F(m): solution below m, so $A_{hi} = m$

7 Step 6: calibrating the APL gain

The gain a sets how strong APL is. It is chosen so that, with APL intact, 10% of KCs respond on average over odors, half the blocked level. More gain means more inhibition and fewer KCs responding, so the mean fraction falls steadily as a grows. The code (function `calib`) starts at $a = 1$ and doubles it until the fraction drops to 10% or below. It then bisects 50 times between the last two values. Every evaluation solves the fixed point of Step 5 for all odors.

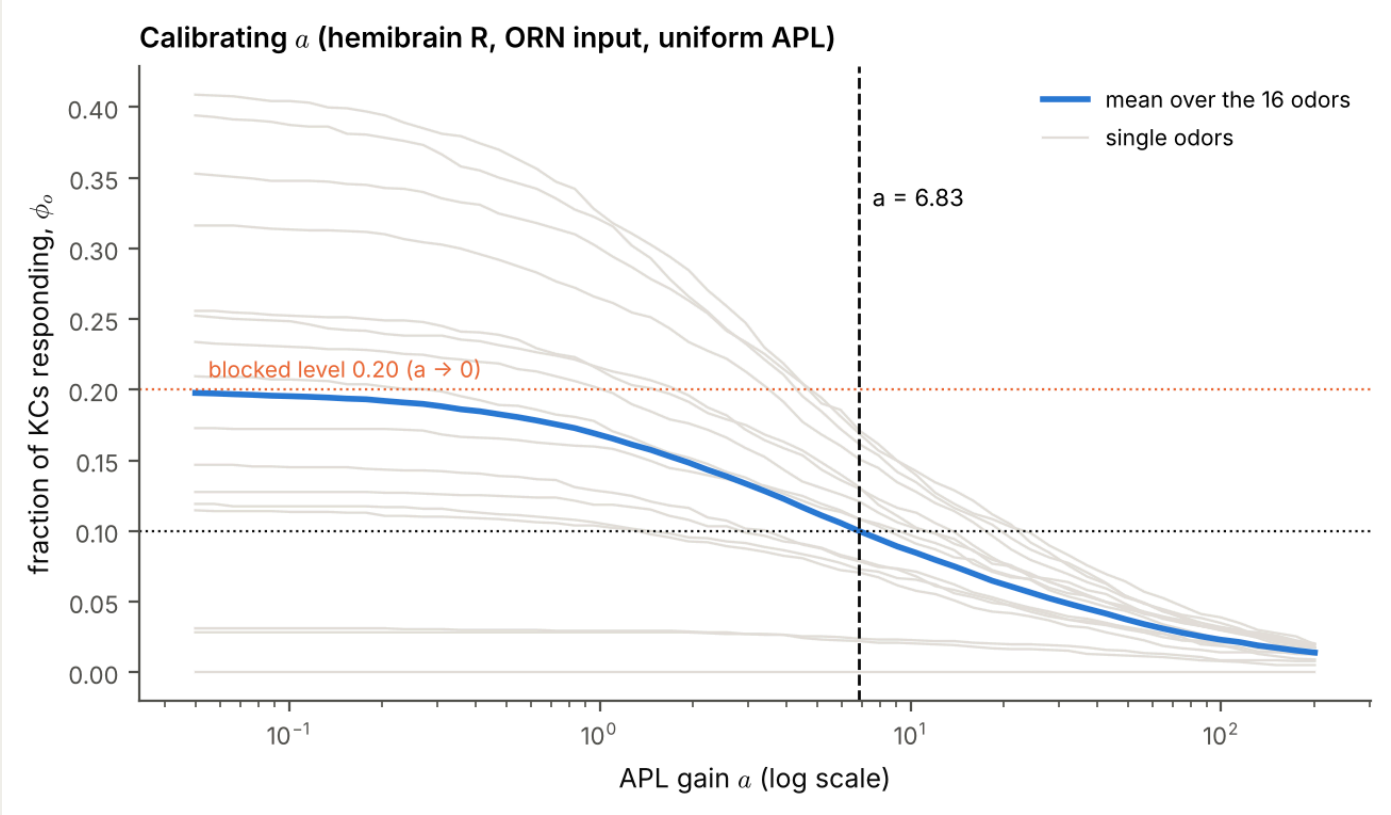


Figure 8. Calibrating the APL gain. The fraction of KCs responding with APL intact as the gain a grows (log axis). Thin grey lines are single odors, and the blue line is their mean. The calibrated gain (dashed) is where the mean reaches 0.10 (dotted black). At $a \to 0$ the model is the blocked model, with a mean of 0.20 (dotted orange).

Table 5. Thresholds and calibrated gains (20% → 10%), for every input. U and C give a (subtractive); P0, P and L give b (divisive, on PN boutons or claws); M0, M and MC give the common scale s (Section 9). The gains are not comparable across columns, because each version uses a different form and normalization. Cs and Cc are calibrated in the same way as C and are not shown. The inputs are described in Section 4.

input	mushroom body	θ	U	C	P0	P	L	M0	M	MC
ORN	hemibrain R	0.352	6.83	11.2	25.6	24.8	26.9	0.975	0.97	0.981
ORN	male CNS L	0.366	7.77	8.93	27.9	27.3	27.6	0.974	0.975	0.959
ORN	male CNS R	0.369	6.88	9.49	24.4	23.9	25	0.972	0.965	0.972
GH146	hemibrain R	0.273	6.28	9.18	34	38.2	38.1	0.962	0.986	0.971
GH146	male CNS L	0.274	5.96	6.41	31.3	34.6	34.4	0.964	0.983	0.951
GH146	male CNS R	0.287	6.59	8.65	33.8	38.4	36.7	0.964	0.973	0.944
ORN-PN	hemibrain R	0.061	6.54	10.3	149	148	160	0.966	0.97	0.973
ORN-PN	male CNS L	0.064	7.35	8.3	161	153	162	0.967	0.955	0.938

input	mushroom body	θ	U	C	P0	P	L	M0	M	MC
ORN-PN	male CNS R	0.064	6.64	8.93	141	144	146	0.969	0.97	0.967
ORN neg	hemibrain R	0.351	6.86	11.2	25.7	25.1	27	0.974	0.967	0.974
ORN neg	male CNS L	0.366	7.76	8.94	28	27.1	27.9	0.973	0.974	0.958
ORN neg	male CNS R	0.369	6.83	9.44	24.4	24	24.9	0.972	0.972	0.972
GH146 neg	hemibrain R	0.271	6.27	9.22	34.5	38.7	38.6	0.963	0.998	0.97
GH146 neg	male CNS L	0.273	5.99	6.47	31.4	34.7	34.7	0.964	0.972	0.947
GH146 neg	male CNS R	0.285	6.58	8.63	34.4	38.9	37.1	0.96	0.967	0.943

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8 Step 7: what is read out

Three quantities are read out for every mushroom body, version and input table:

- **Fraction responding per odor**, ϕ_o, blocked and intact.
- **Odor-pair similarity** S_{ab}: the cosine similarity of the two odors' KC population response vectors $\mathbf{y}_a = (y_{1a}, \ldots, y_{Na})$ and $\mathbf{y}_b$ (Equation 9). It is the same measure as Yang's RDMs (which hold $1 - S$). Because responses are ≥ 0, S runs from 0 (no KC responds to both) to 1 (the same pattern up to a scale factor). It is undefined (0/0) when an odor drives no KC.
- **The effect of blocking APL** on a pair, $\Delta S_{ab} = S_{ab}^{\text{blocked}} - S_{ab}^{\text{intact}}$ (Equation 10). It is computed per mushroom body and then averaged over the three.

$$S_{ab} \;=\; \cos(\mathbf{y}_a, \mathbf{y}_b) \;=\; \frac{\sum_k y_{ka}\, y_{kb}}{\sqrt{\sum_k y_{ka}^2}\,\sqrt{\sum_k y_{kb}^2}} \tag{9}$$

$$\Delta S_{ab} \;=\; S_{ab}^{\text{blocked}} - S_{ab}^{\text{intact}} \tag{10}$$

Every odor is kept, including ms. The earlier version of this report left out odors that drove fewer than 0.5% of KCs. Now a similarity is only missing where it cannot be computed. This happens for one odor, ms with ORN input: with APL blocked it drives 0 KCs in hemibrain R, 0 KCs in male CNS L, 1 KCs in male CNS R (intact: 0, 0, 1). Its ORN responses are weak (largest 0.58, against 1.10 for the median odor), and with one threshold for all odors almost no KC reaches it. With GH146 input ms drives 55, 63, 48 KCs, so its pairs are defined there. Averages over the three mushroom bodies use those where the value is defined. In the heat maps, cells that rest on fewer than three mushroom bodies, or on an odor that drives fewer than 10 KCs in some mushroom body, are **hatched**; grey means undefined in all three.

Correlations. Every "r" in this report is a **Pearson** correlation; rank (Spearman) correlations are written ρ.

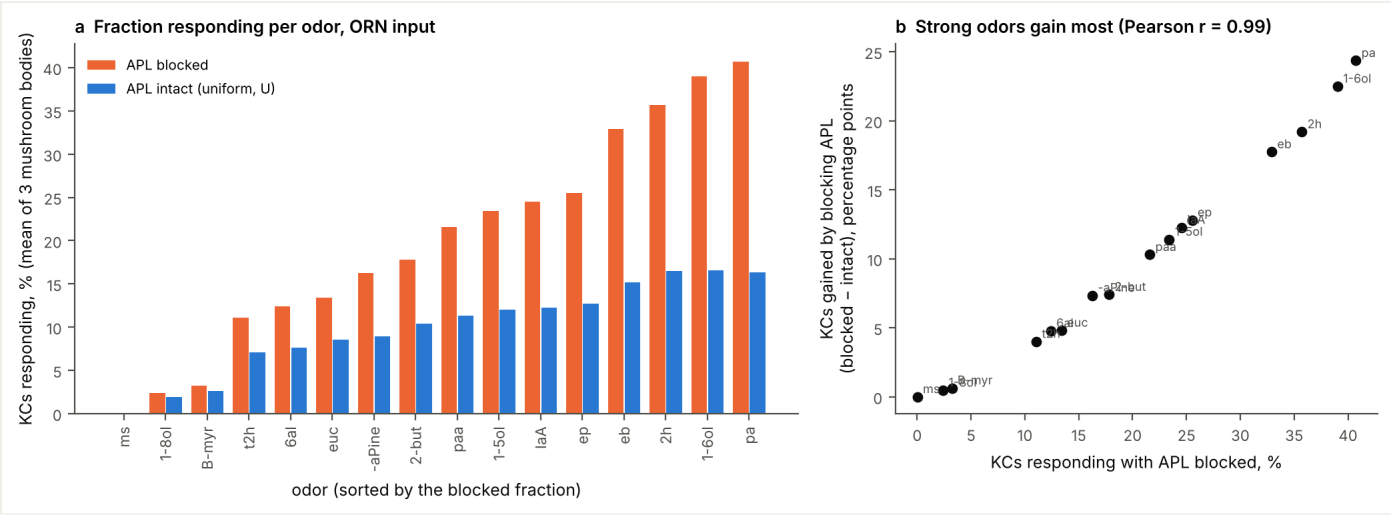


Figure 9. Per-odor fractions responding, blocked and intact. a, Percentage of KCs responding to each odor with APL blocked (orange) and intact (blue, uniform APL), averaged over the three mushroom bodies (ORN input, 20% → 10%). ms is at 0. **b**, One point per odor. The horizontal axis is the blocked percentage (odor strength). The vertical axis is how many percentage points blocking APL adds. Strong odors recruit more APL, so removing APL adds the most KCs for them.

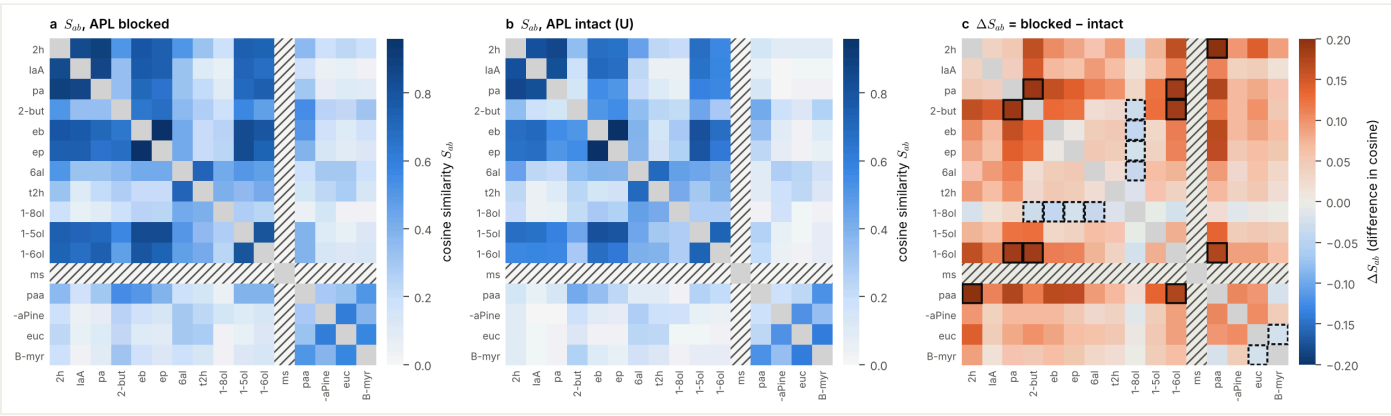


Figure 10. Odor-pair similarity with APL blocked and intact (uniform APL), and the change. Odor-by-odor matrices, mean of the three mushroom bodies, ORN input, uniform APL (U). **a**, **b**, Cosine similarity of the KC population responses with APL blocked and intact, on one shared scale (white 0, dark blue the highest value). **c**, Their difference, ΔS_{ab}, on a diverging scale: orange = more similar when APL is blocked, blue = less similar. **The average change in similarity with minus without inhibition is +0.061** (sd 0.057, range −0.04 to +0.20, 120 pairs). Solid outlines: the 5 pairs that become most similar when APL is blocked (2h–paa (+0.201), pa–2-but (+0.189), 2-but–1-6ol (+0.184), pa–1-6ol (+0.183), 1-6ol–paa (+0.174)); dashed outlines: the 5 that change least or become less similar (eb–1-8ol (−0.039), 6al–1-8ol (−0.036), 2-but–1-8ol (−0.031), euc–B-myr (−0.026), ep–1-8ol (−0.026)). Hatched: ms (0, 0, 1 KCs with APL blocked in hemibrain R, male CNS L, male CNS R). Grey: diagonal or undefined (Section 8).

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9 Step 8: versions that use APL's connectome wiring

Each version replaces part of the uniform APL with a connectome measurement. All of them keep the same blocked responses and the same θ, and all are calibrated to the same 10%. The versions therefore differ only in **where** the inhibition acts and **how it is weighted**. Table 6 summarizes them.

Table 6. The APL versions of the prediction run. a_U, a_C, b_{P0} and b_P are the calibrated gains of U, C, P0 and P. In M0, M and MC only the common scale s is calibrated. Report 07 adds two feed-forward versions (F0, F), where APL activity comes from the PN input instead of the KCs.

version	KC response y_{ko}	APL activity A_o	weights from the connectome
U, uniform	$[x_{ko} - \theta - aA_o]_+$	$\frac{1}{N}\sum_k y_{ko}$	none ($u = v = 1$)

version	KC response y_{ko}	APL activity A_o	weights from the connectome
C, per-KC counts	$[x_{ko} - \theta - a u_k A_o]_+$	$\frac{1}{N}\sum_k v_k y_{ko}$	u_k = APL→KC, v_k = KC→APL synapses of KC k in the calyx, each ÷ its mean
Cs, shuffled control	as C	as C	(u_k, v_k) pairs shuffled among KCs of the same class (αβ, α′β′, γ)
Cc, class means	as C	as C	u_k, v_k replaced by the mean of the KC's class
P0, boutons, all alike	$[\sum_g W_{kg} \frac{r_{go}}{1+bA_o} - \theta]_+$	$\frac{1}{N}\sum_k y_{ko}$	none
P, boutons per glomerulus	$[\sum_g W_{kg} \frac{r_{go}}{1+b\,p_g A_o} - \theta]_+$	as P0	p_g = APL→PN per PN→KC synapse of glomerulus g
L, claws per glomerulus	$[\sum_g W_{kg} \frac{r_{go}}{1+b\,q_g A_o} - \theta]_+$	as P0	q_g = on-claw APL→KC per claw PN synapse of g
M0, U + boutons at real share	$[\sum_g W_{kg} \frac{r_{go}}{1+sfb_{P0}A_o} - \theta - s(1-f)a_U A_o]_+$	as U	f = boutons' share of APL's calyx output
M, U + boutons per glomerulus	as M0 with p_g in the bouton term	as U	f and p_g
MC, C + boutons per glomerulus (new)	$[\sum_g W_{kg} \frac{r_{go}}{1+sfb_P\,p_g A_o} - \theta - s(1-f)a_C\,u_k A_o]_+$	as C	u_k, v_k, p_g and f

9.1 C: APL synapses per KC

u_k is KC k's number of APL→KC synapses in the calyx, divided by the mean over the mushroom body's KCs. v_k is the same for KC→APL synapses (calyx only since 2026-10-21, at Yang's request). A KC with $u_k = 2$ receives twice the average inhibition for the same APL activity, and a KC with $v_k = 2$ counts twice as much toward APL's activity. APL is still one unit with one activity per odor. The two controls ask whether it matters **which** KC has which counts (Cs keeps each KC class's distribution but shuffles it) or only the class averages (Cc).

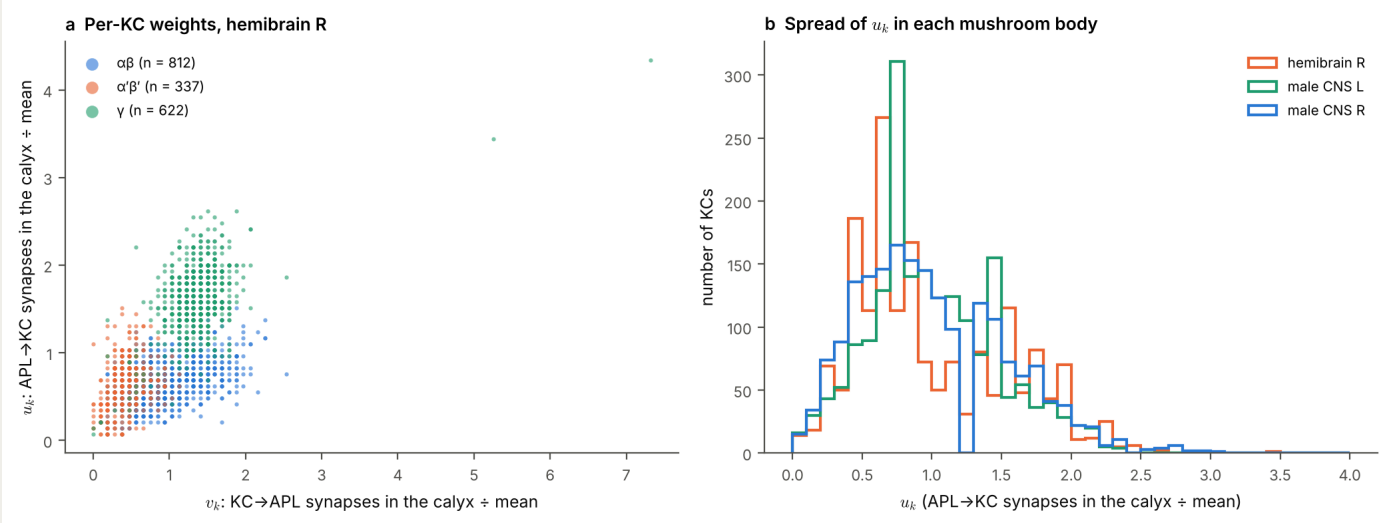


Figure 11. The per-KC weights used by version C. a, Each point is one hemibrain KC. Horizontal: KC→APL synapses in the calyx divided by the mushroom body mean (v_k). Vertical: APL→KC synapses divided by their mean (u_k). Colors are KC classes. **b**, Histograms of u_k in each mushroom body. Uniform APL corresponds to every KC sitting at (1, 1).

9.2 P and L: APL synapses per glomerulus, and why they are counted per synapse

In P and L, inhibition **divides** a glomerulus's input instead of subtracting from the KC, because APL→PN synapses act presynaptically on the PN boutons and scale down what the boutons pass to all their KCs. The input that glomerulus g sends to every KC is divided by $1 + b\,\tilde{p}_g\,A_o$. So $\tilde{p}_g$ has to answer one question: *how strongly is each unit of g's input inhibited?*

Raw APL synapse counts answer a different question, because big glomeruli have more of everything. In the hemibrain, DC1 has 6,126 PN→KC synapses and VA1v 1,968. If both had the same number of APL→PN synapses per PN→KC synapse, DC1 would still have about 3 times as many in total. The size of a glomerulus's input already enters the model through W_{kg}. Using raw APL counts would count size a second time and make big glomeruli look strongly inhibited simply because they are big. Dividing by the excitatory synapses removes that:

- p_g (version P) is the number of APL→PN synapses on glomerulus g's boutons per PN→KC synapse of those boutons: the number of APL synapses that each PN→KC synapse of g "sees".
- q_g (version L) is the same for the APL→KC synapses that sit on the claws of glomerulus g, divided by the PN synapses on those claws. A glomerulus with many large claws is then not counted as more inhibited only because it has more claw membrane.

Table 7. Raw counts and the per-synapse weights, hemibrain (corrected calyx, new claws). The raw counts (columns 4 and 7) mostly follow the glomerulus's size (columns 3 and 6). The ratios (columns 5 and 8) are what P and L use, after scaling to a claw-input-weighted mean of 1 (Equation 11).

glomerulus	PNs	PN→KC synapses	APL→PN synapses (boutons)	p_g = APL→PN ÷ PN→KC	PN synapses on its claws	APL→KC synapses on its claws	q_g = APL→KC ÷ claw PN synapses
DC1	1	6,126	57	0.009	6,078	531	0.087
DM6	4	5,251	90	0.017	5,211	345	0.066
DM2	2	4,954	102	0.021	4,899	394	0.080
DA1	7	4,784	71	0.015	4,734	546	0.115
DL5	1	2,551	35	0.014	2,521	201	0.080
VM2	2	2,001	13	0.006	1,972	146	0.074
VA1v	3	1,968	33	0.017	1,965	174	0.089
DA2	5	2,083	18	0.009	2,045	185	0.090

Table 8. Why per synapse, in a mock example. Two made-up glomeruli. Raw counts would rank A as the more inhibited glomerulus; per synapse, B is.

	glomerulus A (big)	glomerulus B (small)	what it means
PN→KC synapses	1,000	200	A sends 5 times more input to KCs
APL→PN synapses on the boutons	100	40	A receives 2.5 times more APL synapses in total
p_g (per PN→KC synapse)	0.10	0.20	but each unit of B's output is inhibited twice as strongly
a KC with half its input from each	$\frac{0.5\,r_A}{1+0.10\,bA_o}$	$\frac{0.5\,r_B}{1+0.20\,bA_o}$	B's half of the input is divided more

Within each mushroom body, the weights are scaled so that their claw-input-weighted mean is 1:

$$\tilde{p}_g = \frac{p_g}{\left(\sum_{g'} w_{g'}\, p_{g'}\right) / \sum_{g'} w_{g'}}, \qquad w_g = \sum_k W_{kg} \tag{11}$$

sums over glomeruli g' that have a value; glomeruli without a value get $\tilde{p}_g = 1$

With this scaling, P and P0 put the same total inhibition on the average claw input, and only its distribution across glomeruli differs. Report 04 shows p_g and q_g across glomeruli and connectomes.

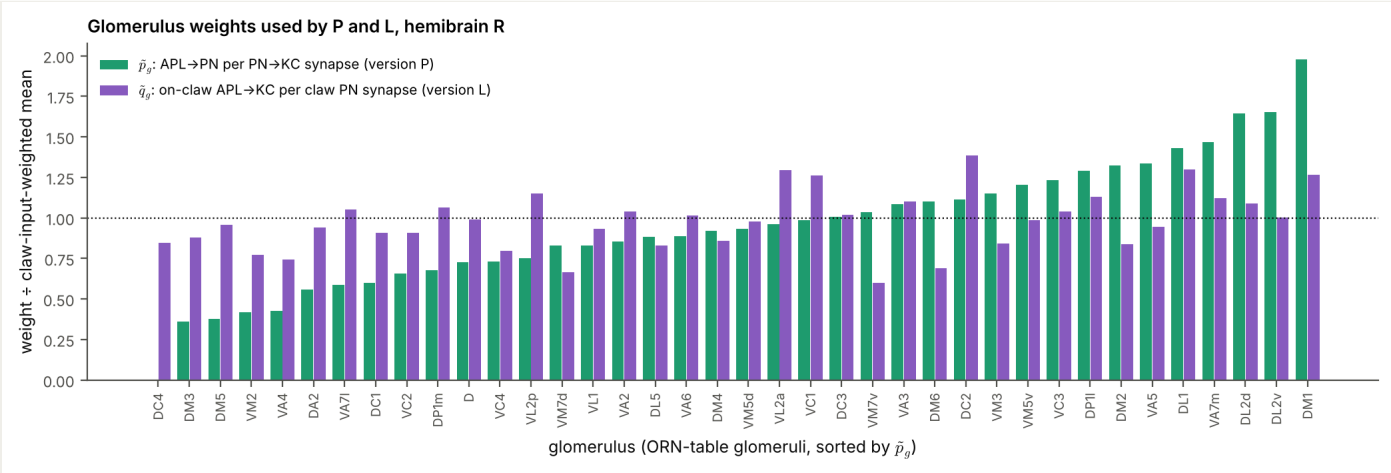


Figure 12. Per-glomerulus weights after scaling. For each glomerulus of the ORN table (hemibrain R), the bouton weight $\tilde{p}_g$ (green) and claw weight $\tilde{q}_g$ (purple), each divided by its claw-input-weighted mean (Equation 11), so the dotted line at 1 is the average. A glomerulus at 2 gets twice the average inhibition per unit of input in that version.

9.3 M0, M and MC: inhibition on both KCs and boutons

APL makes synapses onto both KCs and PN boutons in the calyx. In M0, M and MC, only a share f of the inhibition goes to boutons, the boutons' share of APL's calyx output synapses: f = APL→PN ÷ (APL→PN + APL→KC), 0.1 in hemibrain and 0.103 in male CNS. The rest acts on the KCs.

- **M0**: uniform on the KCs (as U) and on all boutons alike (as P0). KC gain $(1-f)\,a_U$, bouton gain $f\,b_{P0}$.
- **M**: as M0, but the bouton term uses the per-glomerulus weights $\tilde{p}_g$ (as P).
- **MC** (new, item 5 of the revision): inhibition on both, each with its connectome weights. The KC term is that of C (per-KC u_k, and APL's activity weighted by v_k), with gain $(1-f)\,a_C$; the bouton term is that of P, with gain $f\,b_P$ and $\tilde{p}_g$.

In all three, both gains are scaled by one common factor s, the only number calibrated, so that 10% of KCs respond.

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10 Results of the prediction run

All results are for the main setting (20% of KCs respond with APL blocked, 10% intact), averaged over the three mushroom bodies. Table 9 says exactly what each number is: what is correlated with what, and over which items. "With ORN input" only says which response table drove the model.

Table 9. Every number in the Results, and what it is. ϕ_o is the fraction of KCs responding to odor o. "MB" = mushroom body. Pairs with ms count where they are defined (Section 8). r = Pearson correlation, ρ = Spearman (rank) correlation.

statement	what is computed	over what	value (ORN / GH146)

statement	what is computed	over what	value (ORN / GH146)
KCs gained by blocking APL track odor strength	Pearson r between $\phi_o^{\text{blocked}} - \phi_o^{\text{intact,U}}$ (KCs gained) and ϕ_o^{blocked} (odor strength); each fraction averaged over the 3 MBs first	16 odors (ORN) / 13 (GH146)	0.991 / 0.993
and so do all versions	Pearson r between the KCs gained under U and under each other version; the lowest version is given	odors	0.998 (L) / 0.997 (P)
mean ΔS	mean of ΔS_{ab} (Equation 10; cosine), each pair averaged over the MBs where it is defined	odor pairs	+0.061 (120 pairs) / +0.033 (78 pairs)
similarity rises most for pairs of strong odors	Spearman ρ between pair strength $\sqrt{\phi_a^{\text{blocked}}\,\phi_b^{\text{blocked}}}$ and ΔS_{ab} (U); Figure 13	odor pairs	0.76 / 0.66
same pattern in all three MBs	Pearson r between two MBs' ΔS (U) over pairs, for each of the 3 MB pairs; the median of the 3	odor pairs	0.94 / 0.87
per-KC counts (C) change ΔS	mean over pairs and MBs of $\lvert\Delta S_{ab}^{C} - \Delta S_{ab}^{U}\rvert$	pairs × 3 MBs	0.016 / 0.015
shuffled counts do much the same	Pearson r between the maps $\Delta S^{C} - \Delta S^{U}$ and $\Delta S^{Cs} - \Delta S^{U}$ (MB means)	odor pairs	0.63 / 0.72
what is left differs between MBs	as the "same pattern" row, for $\Delta S^{C} - \Delta S^{Cs}$	odor pairs	0.16 / 0.05
P against P0	median Pearson r between MBs of $\Delta S^{P} - \Delta S^{P0}$, and mean $\lvert\Delta S^{P} - \Delta S^{P0}\rvert$	pairs (× 3 MBs)	r 0.85, mean |diff| 0.026 / r 0.85, mean |diff| 0.032
L against P0	the same for $\Delta S^{L} - \Delta S^{P0}$	pairs (× 3 MBs)	r 0.85, mean |diff| 0.011 / r 0.67, mean |diff| 0.011
M against M0	the same for $\Delta S^{M} - \Delta S^{M0}$	pairs (× 3 MBs)	r 0.89, mean |diff| 0.005 / r 0.83, mean |diff| 0.005
MC against M and C	the same for $\Delta S^{MC} - \Delta S^{M}$; then for $\Delta S^{MC} - \Delta S^{C}$	pairs (× 3 MBs)	r 0.24, mean |diff| 0.015 / r 0.25, mean |diff| 0.013; r 0.87, mean |diff| 0.005 / r 0.69, mean |diff| 0.006

10.1 Similarity rises most for pairs of strong odors

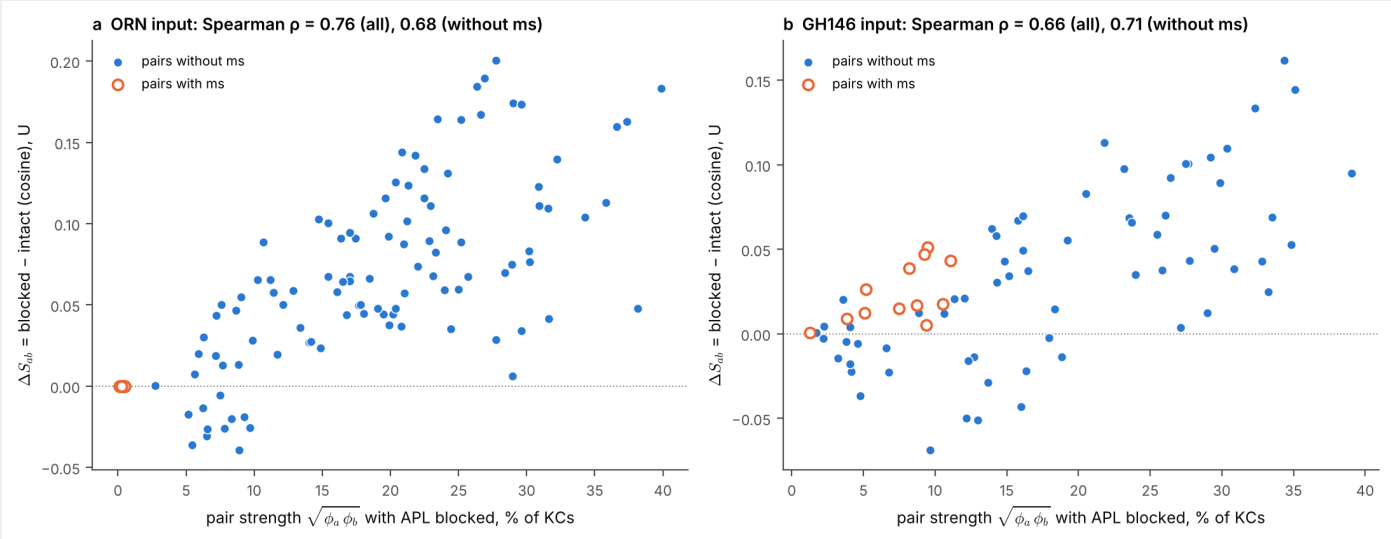


Figure 13. Similarity change against pair strength. Each point is one odor pair (uniform APL, mean of the three mushroom bodies, 20% → 10%). Horizontal: the geometric mean of the two odors' fractions of KCs responding with APL blocked. Vertical: how much more similar the pair is with APL blocked. Open orange circles are the pairs with ms. With ORN input ms drives no KC in two mushroom bodies, so its pairs come from male CNS R only, where it drives 1 KC; they sit near pair strength 0.

Pairs of strongly driven odors gain the most similarity (Spearman ρ = 0.76 with ORN input, 0.66 with GH146). Strong odors drive many KCs, recruit more APL, and so lose more of their responses to APL; when APL is blocked, their overlapping KCs come back and their patterns converge.

10.2 Every version: similarity with and without inhibition, and its difference from U

Table 10. The average change in similarity (blocked – intact) in every version, and how each version differs from U. Mean ΔS: mean over odor pairs of the MB-averaged ΔS_{ab}, with its smallest and largest pair. r with U: Pearson r between the version's and U's MB-averaged ΔS over pairs. Mean $|\Delta S - \Delta S^U|$: over pairs and MBs. The blocked responses are the same in every version, so $\Delta S - \Delta S^U = S^{\text{intact},U} - S^{\text{intact}}$.

version	ORN: mean ΔS (range)	ORN: r with U	ORN: mean $\lvert\Delta S - \Delta S^U\rvert$	GH146: mean ΔS (range)	GH146: r with U	GH146: mean $\lvert\Delta S - \Delta S^U\rvert$
U: uniform APL	+0.061 (−0.04 to +0.20)	—	—	+0.033 (−0.07 to +0.16)	—	—
C: per-KC APL synapse counts	+0.046 (−0.07 to +0.17)	0.99	0.016	+0.019 (−0.07 to +0.12)	0.98	0.015
Cs: per-KC counts shuffled within KC class	+0.048 (−0.04 to +0.16)	0.99	0.018	+0.023 (−0.06 to +0.13)	0.98	0.012
Cc: KC-class means of the counts	+0.054 (−0.04 to +0.18)	1.00	0.009	+0.028 (−0.06 to +0.14)	0.99	0.007
P0: all inhibition on PN boutons, every glomerulus alike	+0.062 (−0.04 to +0.20)	1.00	0.002	+0.034 (−0.08 to +0.16)	1.00	0.003

version	ORN: mean ΔS (range)	ORN: r with U	ORN: mean $\lvert\Delta S - \Delta S^U\rvert$	GH146: mean ΔS (range)	GH146: r with U	GH146: mean $\lvert\Delta S - \Delta S^U\rvert$
P: all inhibition on PN boutons, per glomerulus ($\tilde{p}_g$)	+0.069 (−0.10 to +0.29)	0.94	0.025	+0.046 (−0.15 to +0.23)	0.81	0.033
L: all inhibition on claws, per glomerulus ($\tilde{q}_g$)	+0.059 (−0.05 to +0.21)	0.97	0.011	+0.028 (−0.10 to +0.13)	0.97	0.012
M0: U plus boutons at their real share	+0.061 (−0.04 to +0.20)	1.00	< 0.001	+0.033 (−0.07 to +0.16)	1.00	< 0.001
M: U plus boutons per glomerulus at their real share	+0.062 (−0.05 to +0.21)	1.00	0.005	+0.034 (−0.07 to +0.17)	0.99	0.005
MC: C plus boutons per glomerulus at their real share	+0.050 (−0.07 to +0.18)	0.98	0.014	+0.022 (−0.08 to +0.14)	0.97	0.013

Figures 14–22 show, for each version with ORN input, the intact similarity, the change ΔS (same layout and scale as Figure 10c, with the 5 most and 5 least changed pairs outlined), and the difference from U (diverging scale clipped at ±0.1). Figure 23 shows all versions with GH146 input.

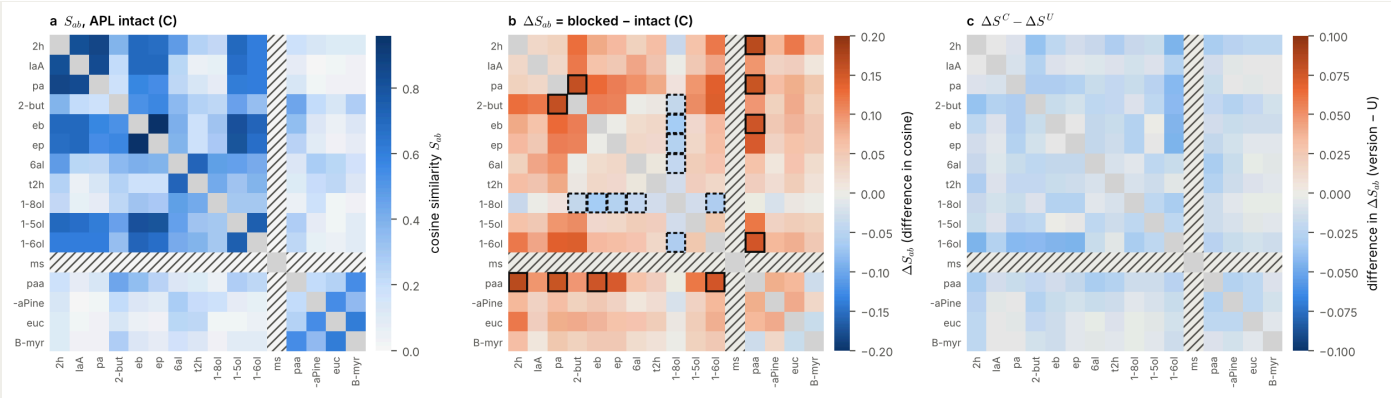


Figure 14. Version C: similarity with APL intact, the change, and the difference from U. Version C (per-KC APL synapse counts), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.046** (U: +0.061; range −0.07 to +0.17). Solid outlines: the 5 pairs that change most (2h–paa (+0.168), pa–2-but (+0.159), pa–paa (+0.156), eb–paa (+0.153), 1-6ol–paa (+0.151)); dashed: the 5 that change least (eb–1-8ol (−0.066), 1-8ol–1-6ol (−0.058), ep–1-8ol (−0.053), 6al–1-8ol (−0.042), 2-but–1-8ol (−0.039)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 0.99 over pairs, mean |difference| 0.016. Hatched: ms (Section 8).

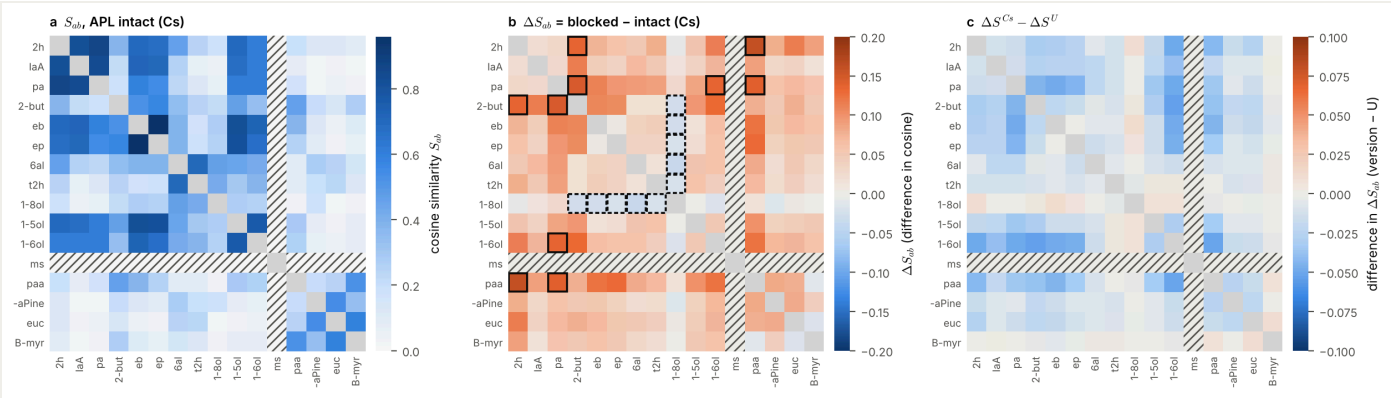


Figure 15. Version Cs: similarity with APL intact, the change, and the difference from U. Version Cs (per-KC counts shuffled within KC class), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.048** (U: +0.061; range −0.04 to +0.16). Solid outlines: the 5 pairs that change most (2h–paa (+0.158), pa–2-but (+0.145), pa–paa (+0.138), pa–1-6ol (+0.134), 2h–2-but (+0.133)); dashed: the 5 that change least (6al–1-8ol (−0.037), 2-but–1-8ol (−0.027), eb–1-8ol (−0.027), t2h–1-8ol (−0.021), ep–1-8ol (−0.020)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 0.99 over pairs, mean |difference| 0.018. Hatched: ms (Section 8).

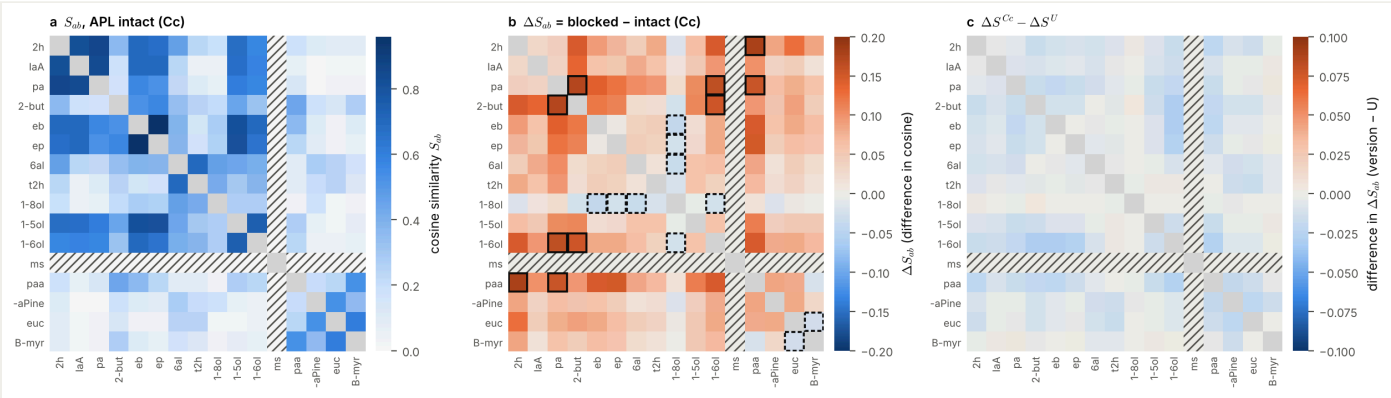


Figure 16. Version Cc: similarity with APL intact, the change, and the difference from U. Version Cc (KC-class means of the counts), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.054** (U: +0.061; range −0.04 to +0.18). Solid outlines: the 5 pairs that change most (2h–paa (+0.179), pa–2-but (+0.172), pa–1-6ol (+0.164), pa–paa (+0.156), 2-but–1-6ol (+0.154)); dashed: the 5 that change least (eb–1-8ol (−0.038), 6al–1-8ol (−0.035), euc–B-myr (−0.028), 1-8ol–1-6ol (−0.026), ep–1-8ol (−0.026)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 1.00 over pairs, mean |difference| 0.009. Hatched: ms (Section 8).

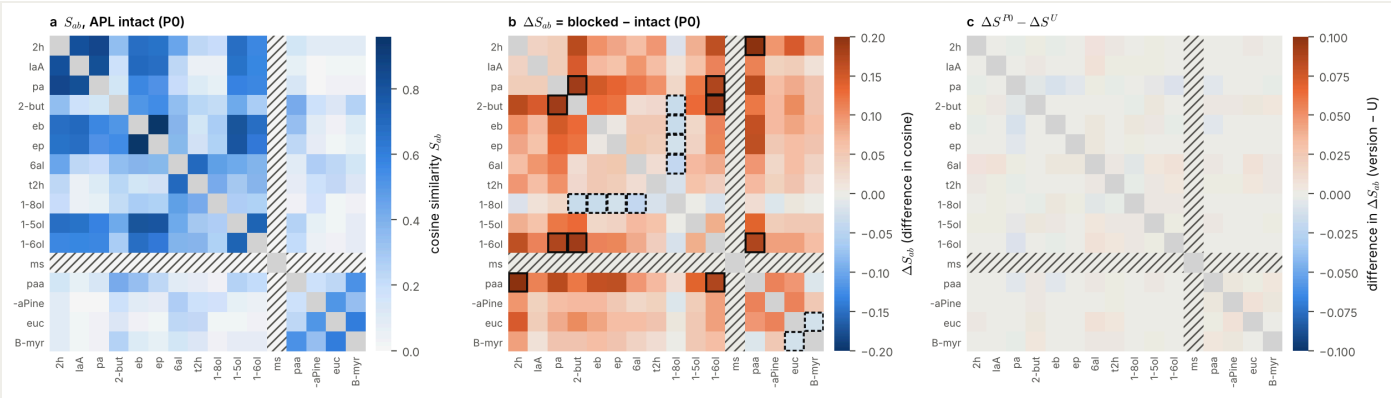


Figure 17. Version P0: similarity with APL intact, the change, and the difference from U. Version P0 (all inhibition on PN boutons, every glomerulus alike), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.062** (U: +0.061; range −0.04 to +0.20). Solid outlines: the 5 pairs that change most (2h–paa (+0.201), 2-but–1-6ol (+0.186), pa–2-but (+0.185), pa–1-6ol (+0.174), 1-6ol–paa (+0.173)); dashed: the 5 that change least (6al–1-8ol (−0.041), eb–1-8ol (−0.035), 2-but–1-8ol (−0.033), euc–B-myr (−0.026), ep–1-8ol (−0.024)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 1.00 over pairs, mean |difference| 0.002. Hatched: ms (Section 8).

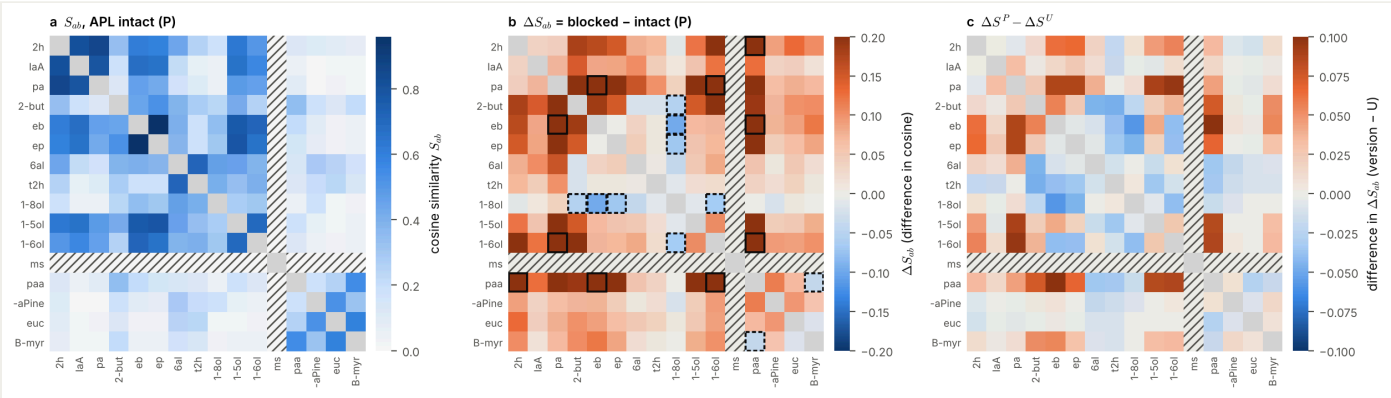


Figure 18. Version P: similarity with APL intact, the change, and the difference from U. Version P (all inhibition on PN boutons, per glomerulus ($\tilde{p}_g$)), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.069** (U: +0.061; range −0.10 to +0.29). Solid outlines: the 5 pairs that change most (eb–paa (+0.292), pa–1-6ol (+0.279), 1-6ol–paa (+0.261), pa–eb (+0.248), 2h–paa (+0.235)); dashed: the 5 that change least (eb–1-8ol (−0.097), ep–1-8ol (−0.071), 1-8ol–1-6ol (−0.066), 2-but–1-8ol (−0.054), paa–B-myr (−0.039)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 0.94 over pairs, mean |difference| 0.025. Hatched: ms (Section 8).

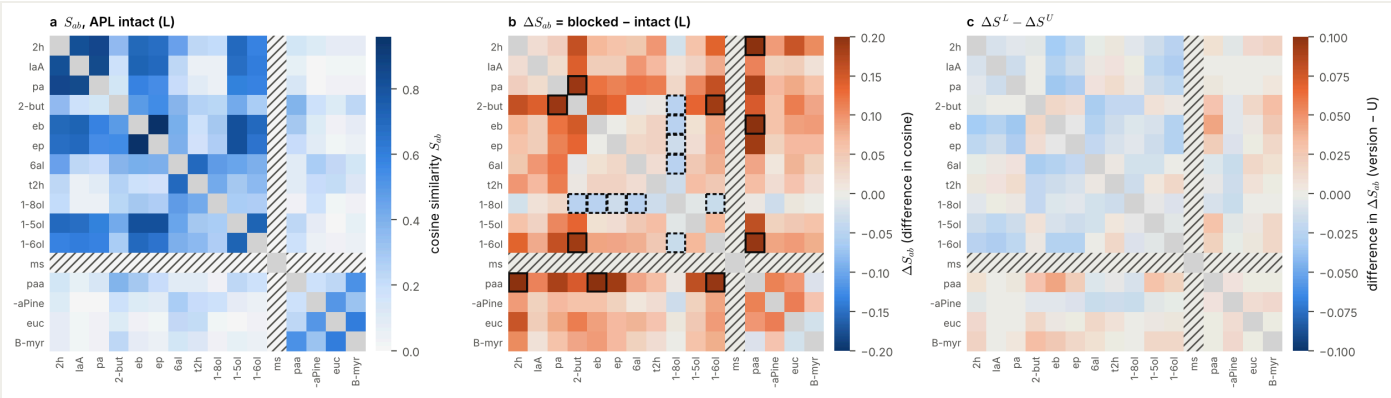


Figure 19. Version L: similarity with APL intact, the change, and the difference from U. Version L (all inhibition on claws, per glomerulus ($\tilde{q}_g$)), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.059** (U: +0.061; range −0.05 to +0.21). Solid outlines: the 5 pairs that change most (eb–paa (+0.210), 2h–paa (+0.209), 1-6ol–paa (+0.193), pa–2-but (+0.190), 2-but–1-6ol (+0.187)); dashed: the 5 that change least (2-but–1-8ol (−0.051), 6al–1-8ol (−0.049), eb–1-8ol (−0.047), 1-8ol–1-6ol (−0.034), ep–1-8ol (−0.032)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 0.97 over pairs, mean |difference| 0.011. Hatched: ms (Section 8).

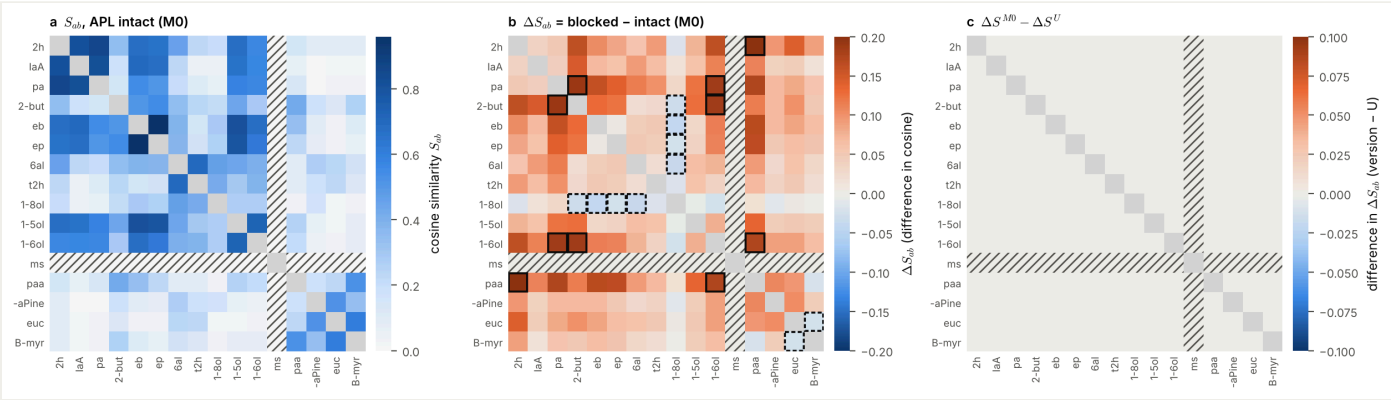


Figure 20. Version M0: similarity with APL intact, the change, and the difference from U. Version M0 (U plus boutons at their real share), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.061** (U: +0.061; range −0.04 to +0.20). Solid outlines: the 5 pairs that change most (2h–paa (+0.201), pa–2-but (+0.189), 2-but–1-6ol (+0.184), pa–1-6ol (+0.183), 1-6ol–paa (+0.174)); dashed: the 5 that change least (eb–1-8ol (−0.039), 6al–1-8ol (−0.036), 2-but–1-8ol (−0.031), euc–B-myr (−0.026), ep–1-8ol (−0.026)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 1.00 over pairs, mean |difference| < 0.001. Hatched: ms (Section 8).

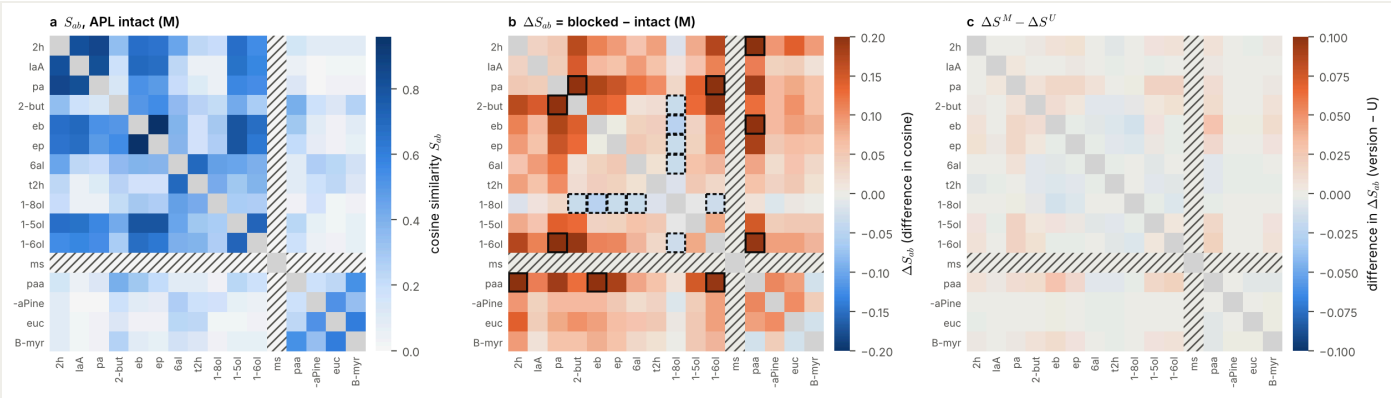


Figure 21. Version M: similarity with APL intact, the change, and the difference from U. Version M (U plus boutons per glomerulus at their real share), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.062** (U: +0.061; range −0.05 to +0.21). Solid outlines: the 5 pairs that change most (2h–paa (+0.208), pa–1-6ol (+0.200), pa–2-but (+0.197), eb–paa (+0.196), 1-6ol–paa (+0.193)); dashed: the 5 that change least (eb–1-8ol (−0.051), 6al–1-8ol (−0.036), 2-but–1-8ol (−0.034), ep–1-8ol (−0.033), 1-8ol–1-6ol (−0.033)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 1.00 over pairs, mean |difference| 0.005. Hatched: ms (Section 8).

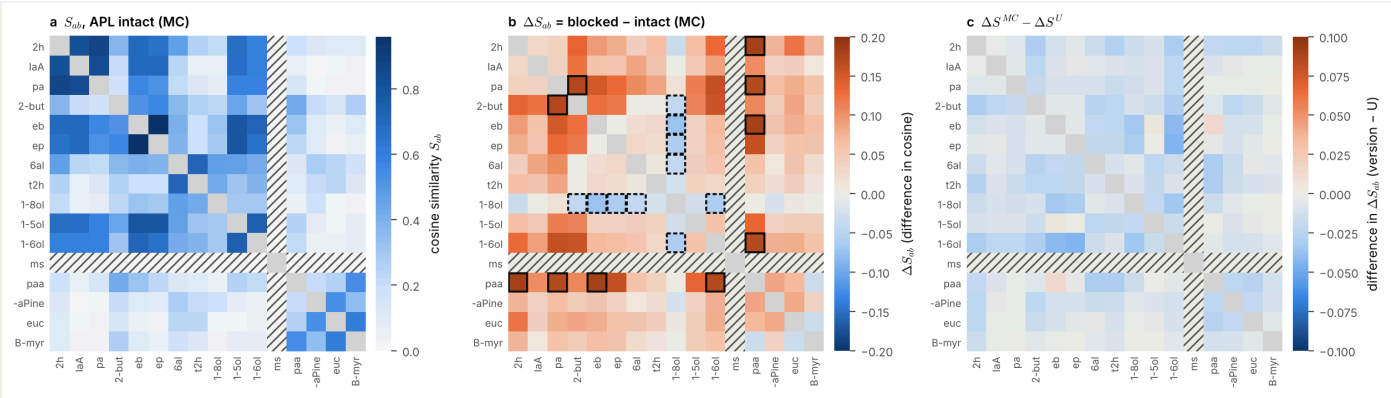


Figure 22. Version MC: similarity with APL intact, the change, and the difference from U. Version MC (C plus boutons per glomerulus at their real share), ORN input, mean of the three mushroom bodies. **a**, Cosine similarity with APL intact. **b**, ΔS_{ab} = blocked − intact. **The average change in similarity with minus without inhibition is +0.050** (U: +0.061; range −0.07 to +0.18). Solid outlines: the 5 pairs that change most (eb–paa (+0.181), 2h–paa (+0.181), pa–2-but (+0.171), 1-6ol–paa (+0.170), pa–paa (+0.170)); dashed: the 5 that change least (eb–1-8ol (−0.073), 1-8ol–1-6ol (−0.061), ep–1-8ol (−0.056), 2-but–1-8ol (−0.041), 6al–1-8ol (−0.040)). **c**, ΔS of this version minus that of U: orange = blocking APL raises the pair's similarity more than under U. Compared with U: Pearson r = 0.98 over pairs, mean |difference| 0.014. Hatched: ms (Section 8).

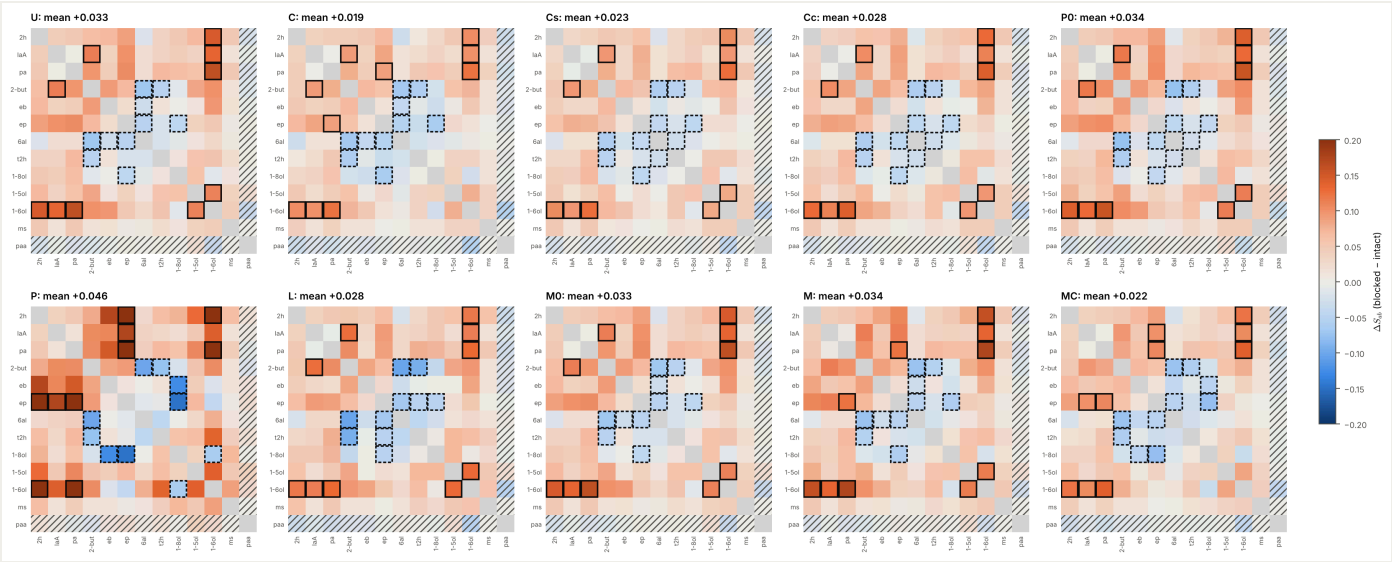


Figure 23. Every version with GH146 input: the change in similarity. ΔS_{ab} = blocked − intact for every version with GH146 input (13 odors), mean of the three mushroom bodies. The panel title gives the average change in similarity with minus without inhibition. Outlines as in Figure 10c (the 5 pairs that change most, solid; least, dashed). Hatched: paa (19, 4, 11 KCs with APL blocked in hemibrain R, male CNS L, male CNS R).

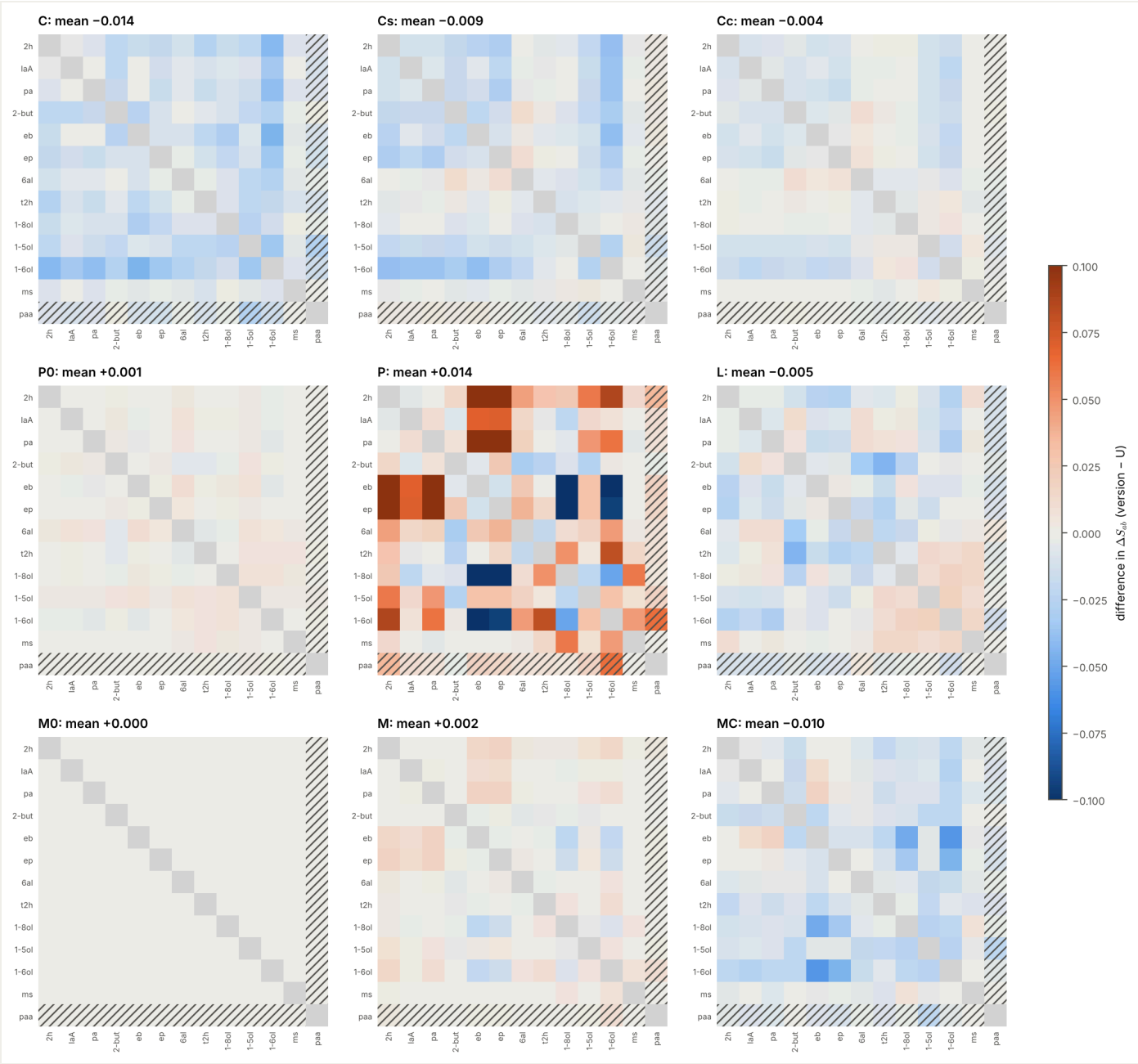


Figure 24. Every version with GH146 input: difference from U. Each version's ΔS_{ab} minus that of U, GH146 input, mean of the three mushroom bodies. The panel title gives the mean difference over pairs. Orange: blocking APL raises the pair's similarity more than under U. Scale clipped at ±0.1.

10.3 Wiring versions against their own references

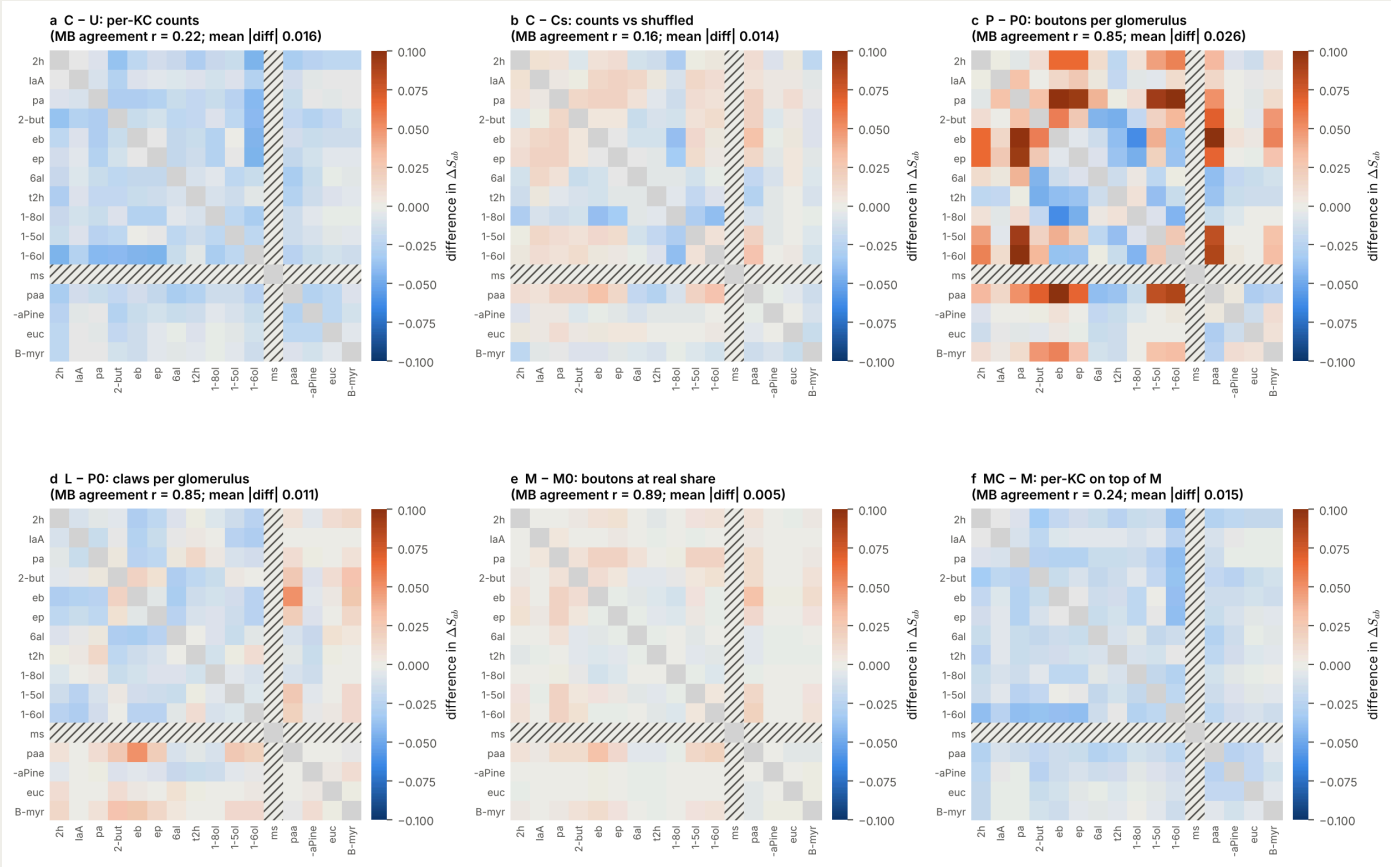


Figure 25. Wiring versions compared with their references. How much each wiring version changes the effect of blocking APL, pair by pair (mean of three mushroom bodies, ORN input). Each cell is ΔS_{ab} of the version minus ΔS_{ab} of its reference. The blocked responses are identical, so this equals the reference's intact similarity minus the version's. Orange: blocking APL raises the pair's similarity more in this version. Blue: less. The scale is clipped at ±0.1. "MB agreement" is the median Pearson r between the three mushroom bodies' maps: a pattern that is the same in all three has r near 1. Hatched: ms (0, 0, 1 KCs with APL blocked in hemibrain R, male CNS L, male CNS R).

10.4 Summary of the prediction run

- **Uniform APL.** The KCs gained by blocking APL track odor strength (Pearson r = 0.99 over odors with ORN input, 0.99 with GH146), and the per-odor gain is nearly the same in all versions (r ≥ 0.997). Similarity rises on average by +0.061 with ORN and +0.033 with GH146, most for pairs of strong odors (Spearman ρ = 0.76 and 0.66). The pattern is the same in all three mushroom bodies (median r 0.94 and 0.87).
- **Per-KC counts (C).** They change ΔS by 0.016 on average (ORN; GH146 0.015). The same counts shuffled within KC class do much the same (maps r 0.63). What is left differs between mushroom bodies (median r 0.16).
- **Boutons per glomerulus (P against P0).** A pattern that is consistent across mushroom bodies (median r 0.85 with ORN, 0.85 with GH146; mean |change| 0.026). The pairs that gain most under P are eb–paa, pa–1-6ol, pa–eb, pa–ep; those that gain least are eb–1-8ol, 2-but–t2h, ep–1-8ol.
- **At the boutons' real share (M against M0)** the same pattern shrinks to a mean |change| of 0.005. **Claws (L against P0)**: mean |change| 0.011 (median r between mushroom bodies 0.85).
- **Both on KCs and boutons with connectome weights (MC).** MC differs from M by 0.015 on average and from C by 0.005 (ORN). Its mean ΔS is in Table 10.

- **Other inputs (Section 11).** The PN transformation gives a mean ΔS of +0.055 (ORN: +0.061). Keeping negative responses gives +0.061 (ORN neg) and +0.033 (GH146 neg; GH146: +0.033).
- **Other weightings (Section 12).** With raw synapse counts, 46% of the responses come from γ KCs (main weighting: 17%). Mean ΔS (U, ORN) is +0.062 with raw counts and +0.055 with equal weight per claw; their maps match the main one with Pearson r = 0.84 and 0.98.

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11 Other inputs: PN-level transformation and negative responses

The same model was run with three more inputs (Section 4): **ORN-PN**, the ORN responses after the Olsen transformation; **ORN neg** and **GH146 neg**, with negative responses kept. Each gets its own threshold and gains. Table 11 compares them with the main inputs.

Table 11. The uniform model with each input. "KCs gained vs odor strength": as in Table 9, row 1. "r with the main input": Pearson r over pairs between the MB-averaged ΔS maps of this input and of ORN (for ORN-PN and ORN neg) or GH146 (for GH146 neg). "r with GH146": the same against GH146, over the pairs of the 13 odors that both have.

input	odors	mean ΔS, U (range)	KCs gained vs odor strength, r	r with the main input's ΔS (U)	r with GH146's ΔS (U), shared odors
ORN	16	+0.061 (−0.04 to +0.20)	0.99	—	0.39 (78 pairs)
ORN-PN	16	+0.055 (−0.04 to +0.18)	0.99	0.99	0.43 (78 pairs)
ORN neg	16	+0.061 (−0.04 to +0.20)	0.99	1.00	0.39 (78 pairs)
GH146	13	+0.033 (−0.07 to +0.16)	0.99	—	—
GH146 neg	13	+0.033 (−0.07 to +0.16)	0.99	1.00	1.00 (78 pairs)

Table 12. Mean change in similarity for every version and input, and the match with GH146. The last two columns: Pearson r between the version's MB-averaged ΔS maps with ORN (or ORN-PN) and with GH146 input, over the pairs of the 13 shared odors. If the PN transformation made the ORN input more PN-like, the last column would be higher than the one before it.

version	mean ΔS, ORN	mean ΔS, GH146	mean ΔS, ORN-PN	mean ΔS, ORN neg	mean ΔS, GH146 neg	ORN vs GH146: r	ORN-PN vs GH146: r
U	+0.061	+0.033	+0.055	+0.061	+0.033	0.39	0.43
C	+0.046	+0.019	+0.041	+0.046	+0.019	0.35	0.39
Cs	+0.048	+0.023	+0.045	+0.047	+0.023	0.34	0.38
Cc	+0.054	+0.028	+0.048	+0.054	+0.028	0.36	0.40
P0	+0.062	+0.034	+0.055	+0.062	+0.034	0.38	0.42
P	+0.069	+0.046	+0.063	+0.069	+0.046	0.54	0.55
L	+0.059	+0.028	+0.052	+0.059	+0.028	0.28	0.32
M0	+0.061	+0.033	+0.055	+0.061	+0.033	0.39	0.43
M	+0.062	+0.034	+0.056	+0.062	+0.034	0.42	0.45
MC	+0.050	+0.022	+0.044	+0.050	+0.022	0.39	0.42

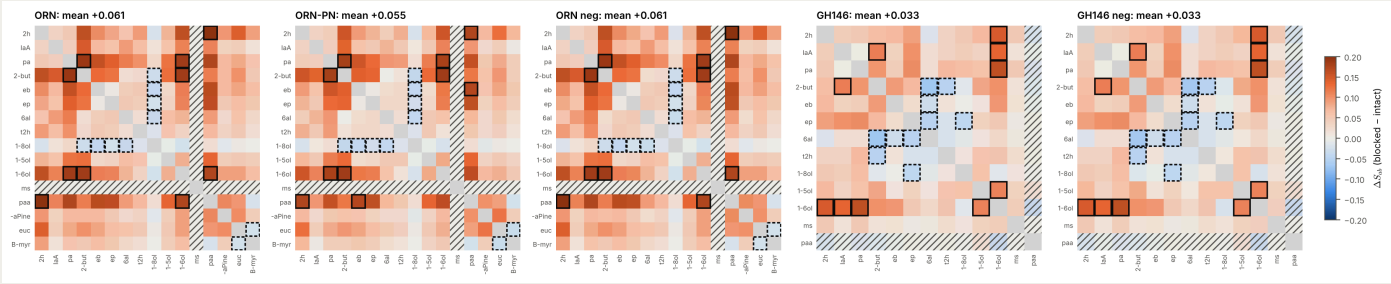


Figure 26. The change in similarity with each input (uniform APL). ΔS_{ab} = blocked − intact for the uniform model with each input, mean of the three mushroom bodies. Outlines: the 5 pairs that change most (solid) and least (dashed). Odors that drive fewer than 10 KCs in some mushroom body are hatched: ORN: ms (0, 0, 1 KCs with APL blocked in hemibrain R, male CNS L, male CNS R); ORN-PN: ms (0, 0, 2 KCs with APL blocked in hemibrain R, male CNS L, male CNS R); ORN neg: ms (0, 0, 1 KCs with APL blocked in hemibrain R, male CNS L, male CNS R); GH146: paa (19, 4, 11 KCs with APL blocked in hemibrain R, male CNS L, male CNS R); GH146 neg: paa (20, 5, 11 KCs with APL blocked in hemibrain R, male CNS L, male CNS R).

- **PN transformation (ORN-PN).** Blocking APL raises similarity a little less (+0.055 against +0.061 with ORN), and the map of which pairs change is nearly the same as with ORN (r = 0.99). Over the shared odors its ΔS map matches GH146's with r = 0.43, against 0.39 for ORN (U; version P, which matches GH146 best, gives 0.55 against 0.54): the transformation moves the model only slightly closer to GH146.
- **ORN and GH146 give different maps.** Even without the transformation, the ORN and GH146 ΔS maps agree only with r = 0.39 over their 78 shared pairs, although both inputs give the same per-odor pattern of KCs gained (Table 11, column 4). Which pairs change most depends on the input table.
- **Negative responses** change almost nothing. They alter the similarity of the KC drives only slightly (mean |change| of the input cosine 0.003 with ORN, 0.012 with GH146) and the KC responses even less (blocked cosine 0.0006 and 0.0016), as the threshold removes small drives anyway. The ΔS maps match those of the clipped inputs with r ≈ 1.00 in every version (Table 12).

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12 Other input weightings

Section 3 explains why W_{kg} is divided by all of the KC's claw PN synapses. Two alternatives were run with the ORN and GH146 inputs:

- W^{raw}: raw synapse counts, $W_{kg} = n_{kg}$, so a KC's drive grows with its number of claw PN synapses. One threshold for all KCs, as before.
- W^{claw}: equal weight per claw, W_{kg} = share of the KC's claws that come from glomerulus g, so a small claw counts as much as a large one.

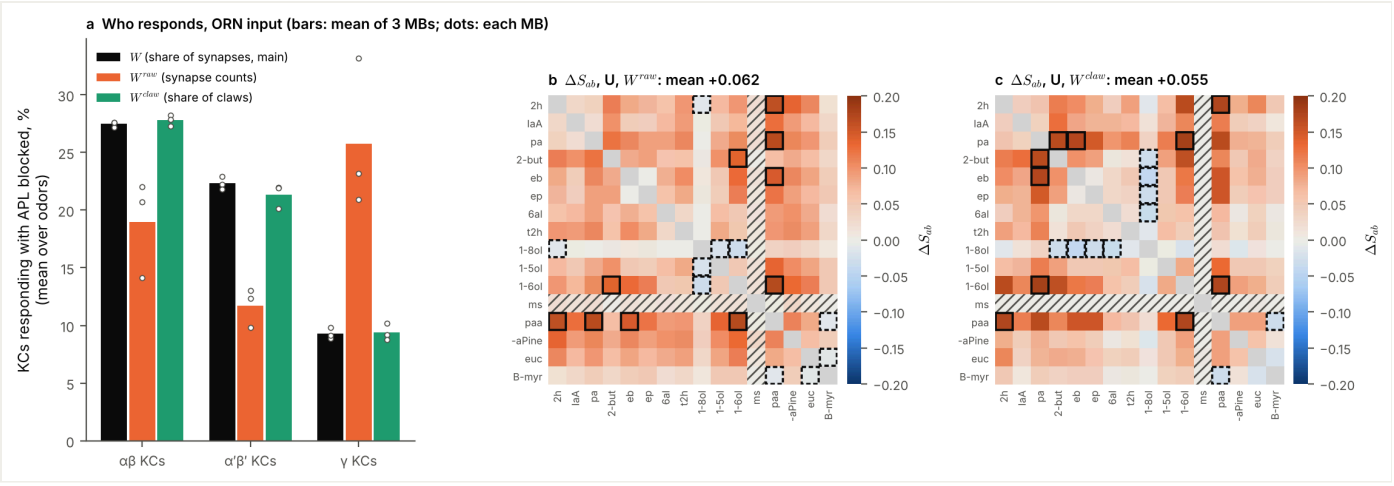


Figure 27. Other input weightings: who responds, and the change in similarity. a, Percentage of KCs of each class that respond with APL blocked (one threshold, 20% of all KCs on average over odors), with the three weightings (ORN input). With raw synapse counts the γ KCs, which have the most claw PN synapses, take most of the responses. **b**, **c**, ΔS_{ab} of the uniform model with the two other weightings (ORN input, mean of three mushroom bodies), with the 5 most (solid) and least (dashed) changed pairs outlined; same scale as the main figure. Raw counts: Hatched: ms (4, 1, 5 KCs with APL blocked in hemibrain R, male CNS L, male CNS R). Equal weight per claw: Hatched: ms (0, 0, 1 KCs with APL blocked in hemibrain R, male CNS L, male CNS R).

Table 13. Mean ΔS for every version with the other weightings, and the match with the main weighting. Each cell: mean ΔS over pairs (MB-averaged), and in brackets the Pearson r over pairs with the same version's ΔS map under the main weighting W.

version	ORN, W^{raw}	ORN, W^{claw}	GH146, W^{raw}	GH146, W^{claw}
U	+0.062 (r 0.84)	+0.055 (r 0.98)	+0.045 (r 0.73)	+0.030 (r 0.98)
C	+0.082 (r 0.66)	+0.041 (r 0.98)	+0.074 (r 0.38)	+0.018 (r 0.99)
Cs	+0.071 (r 0.70)	+0.039 (r 0.96)	+0.069 (r 0.49)	+0.023 (r 0.97)
Cc	+0.076 (r 0.66)	+0.048 (r 0.98)	+0.069 (r 0.52)	+0.025 (r 0.98)
P0	+0.064 (r 0.84)	+0.055 (r 0.97)	+0.048 (r 0.70)	+0.032 (r 0.98)
P	+0.071 (r 0.90)	+0.063 (r 0.99)	+0.053 (r 0.88)	+0.044 (r 0.99)
L	+0.061 (r 0.84)	+0.052 (r 0.97)	+0.045 (r 0.75)	+0.025 (r 0.98)
M0	+0.062 (r 0.84)	+0.055 (r 0.98)	+0.045 (r 0.73)	+0.030 (r 0.98)
M	+0.064 (r 0.86)	+0.056 (r 0.98)	+0.046 (r 0.74)	+0.032 (r 0.98)
MC	+0.081 (r 0.72)	+0.044 (r 0.98)	+0.070 (r 0.47)	+0.021 (r 0.99)

- **Equal weight per claw** gives nearly the same model: γ KCs give 17% of the responses (main weighting 17%; γ KCs are 36% of all KCs), and the ΔS maps match the main ones with r ≥ 0.96.
- **Raw synapse counts** shift the responses to the γ KCs (46% of the responses), because they have the most claw PN synapses (Section 3). With a different set of KCs responding, the maps match the main ones less well (U: r = 0.84 with ORN, 0.73 with GH146), and least for the versions with per-KC counts (C, Cs, Cc, MC).

13 The recordings, for comparison

For direct comparison with the model maps above, Figure 28 repeats the similarity maps of the recordings (Report 06, Figure 3) with the odors in the same order. TNT3c (11 recordings) has APL intact; TNT (8 recordings) has APL blocked. Each recording's similarity is the cosine over its units of the raw responses (FC z-scores), and each condition's map is the mean over its recordings; $1 - S$ is Yang's RDM.

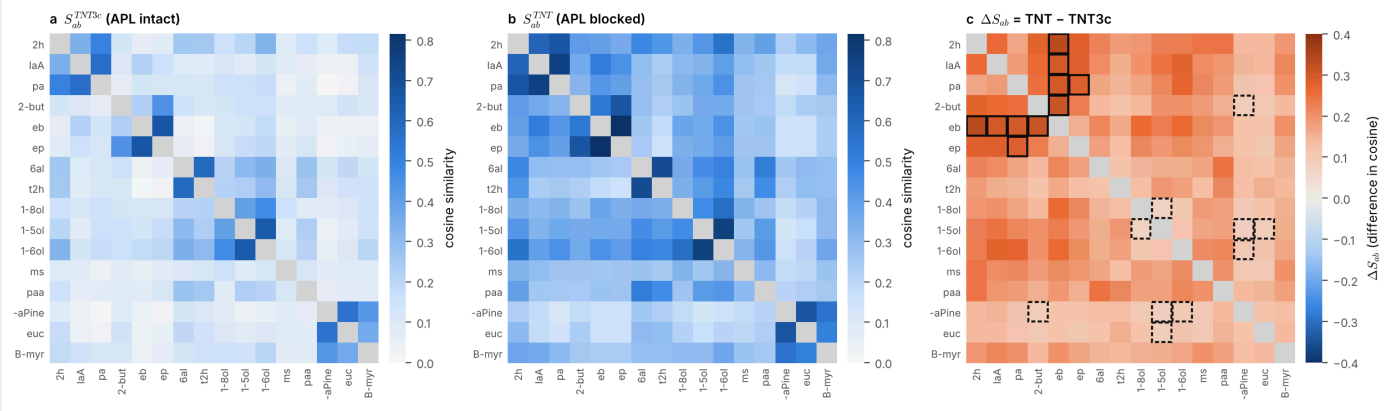


Figure 28. The recordings: odor-pair similarity in TNT3c and TNT, and the change. a, **b**, Mean cosine similarity of the recorded units' responses for every odor pair, in TNT3c and TNT, on one scale (white 0, dark blue the highest value; slightly negative values also show as white). **c**, The change, on a diverging scale (orange = more similar in TNT), clipped at ±0.4, twice the model maps' range. **The average change in similarity is +0.178** (range +0.07 to +0.33; all 120 pairs are more similar in TNT). Solid outlines: the 5 pairs that change most (2h–eb (+0.334), 2-but–eb (+0.310), pa–eb (+0.298), laA–eb (+0.296), pa–ep (+0.284)); dashed: the 5 that change least (1-8ol–1-5ol (+0.074), 1-5ol–-aPine (+0.084), 1-5ol–euc (+0.090), 2-but–-aPine (+0.093), 1-6ol–-aPine (+0.097)).

In the recordings every pair becomes more similar, by +0.18 on average, against +0.061 for the uniform model with ORN input. The model here is set to 20% → 10% of KCs responding for every odor on average, and it runs on another fly's glomerular responses, so its size of effect is not expected to match. Report 07 pins the model to each odor's measured fraction of responding units and compares the two pair by pair.

What this step does not include

One threshold for all KCs, with no differences in excitability between classes, and no spikes or time course.

APL is one unit with one activity per odor in every version. P, L, M and MC weight its output per glomerulus or per KC but do not make its activity local. A local, non-spiking APL is modelled separately (Report 11).

The inputs are ORN or GH146 averages from other flies, not the recorded flies' PNs. The PN transformation's two constants were fitted to GH146; the best fit matches GH146 about as well as the untransformed ORN responses.

With negative responses kept, the divisive versions also shrink negative inputs (Section 4).

The coding levels (20% blocked, 10% intact) are assumptions here. Report 07 replaces them with the recordings.